



Recent advances and future prospects of laser welding technology for polymeric materials: A review

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ABSTRACT

Polymeric materials are extensively used in the production of microfluidic devices, biomedical equipment, automotive parts, and electronic components. Given the limitations of certain manufacturing methods and the complexity of these products, joining polymeric materials becomes essential. Laser welding presents a promising and innovative approach for this task, offering high productivity, the ability to weld intricate shapes, and access to small or hard-to-reach areas. This review explores recent advancements in laser welding technologies, including butt and transmission welding, and the laser systems involved. It also addresses various welding processes such as similar and dissimilar welding, micro-welding, and the welding of polymers to metals. The quality of welds is evaluated based on joint strength and seam width, which can be controlled through parameters like laser power, scan speed, number of scans, and beam diameter. Given the nonlinear response of polymers to heat, determining optimal welding parameters is crucial. Thus, modeling the laser welding process through numerical, analytical, statistical, and artificial intelligence methods is vital for process understanding and optimization. This paper offers a comprehensive review of laser types, welding techniques, and modeling strategies for welding polymeric materials.

Abbreviations

ABS	Acrylonitrile-butadiene-styrene
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ANN	Artificial neural network
ANOVA	Analysis of variance

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ASA	Acrylonitrile-styrene-acrylonitrile
BSS	Beam scanning speed
CB	Carbon black
C-PA6	Carbon-fiber-reinforced polycaprolactam
COC	Cyclic olefin copolymer
CP	Clamping pressure
CFRP	Carbon fiber-reinforced plastic
CNTs	Carbon nanotubes
EDS	Energy dispersive X-ray spectroscopy
ETFE	Ethylene tetrafluoroethylene
FE	Finite element
FV	Finite volume
GA	Genetic algorithm
GF-ABS	Glass-fiber-reinforced acrylonitrile butadiene styrene
GF-ASA	Glass-fiber-reinforced acrylonitrile-styrene-acrylonitrile
GA-PA	Glass-fiber-reinforced polyamide
GF-PA6	Glass-fiber-reinforced polycaprolactam
GF-PP	Glass-fiber-reinforced polypropylene
GFF-MLP	Generalized feed-forward multi-layer perceptrons
HDPE	High-density polyethylene
JS	Joint strength
LBW	Laser butt welding
LDPE	Low-density polyethylene
LIP	Laser incident power
LTW	Laser transmission welding
LWP	Laser welding process
LS	Laser spot
NIH	National Institute of Health

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NIRR	Near-InfraRed region
PBT	Poly(butylene terephthalate)
PC	Polycarbonate
PET	Polyethylene terephthalate
PETG	Polyethylene terephthalate glycol
PEVA	Polyethylene vinyl acetate
PL	Polyester
PMMA	Poly(methylmethacrylate)
PMP	Polymethylpentene
PP	Polypropylene
PS	Polystyrene
PA6	Polycaprolactam
POM	Polyoxymethylene
PVA	Poly(vinyl alcohol)
RBF	Radial basis function
RSM	Response surface methodology
WSW	Weld seam width

1. Introduction

Thermoplastic materials play a significant role in the technological advancements we are experiencing today [1]. This is largely due to their distinctive and attractive properties, including lightweight, design flexibility, moldability, thermal insulation, lower production costs, recyclability, and a wide variety of color options [2]. Once individual

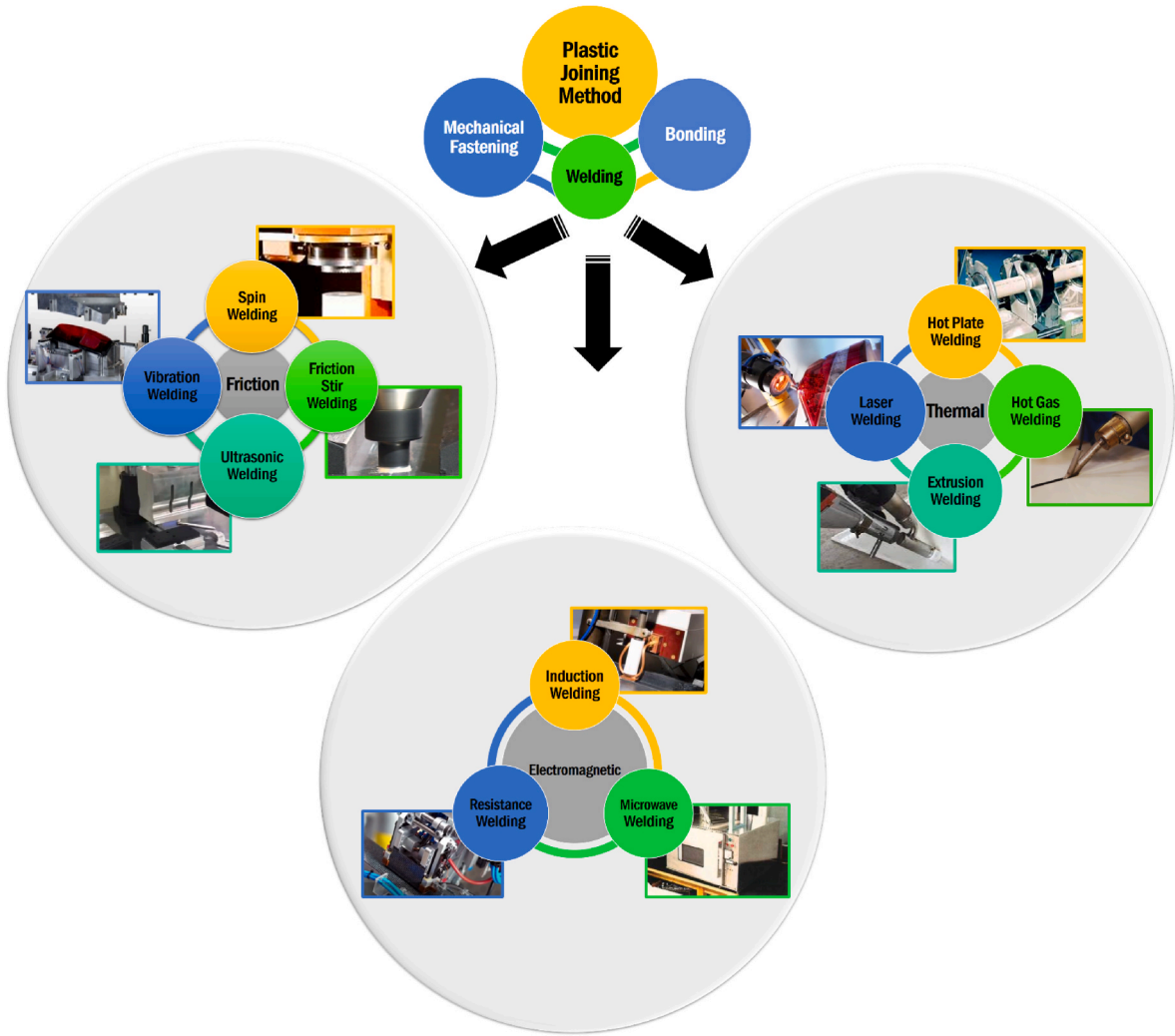


Fig. 1. Plastic joining methods.

plastic components are produced, the joining process is essential, particularly when the final product is large and/or complex [3]. Over the years, several plastic welding techniques such as hot plate welding [4,5], vibration welding [6,7], induction welding [8,9], rotary friction welding [10–13], ultrasonic welding [14,15], microwave welding [16,17], spin rotation welding [18,19], friction stir welding (FSW) [20,21], extrusion welding [22], laser transmission welding (LTW) [23] and resistance welding [24,25] had been developed to ease the assembly procedure and meet the multitudinous industrial needs [26]. Fig. 1 shows the classification of the methods employed to join thermoplastic materials. In addition, Table 1 provides a comparison between laser welding and the other welding processes from different perspectives.

While various welding techniques are employed for joining polymeric materials, each method has its own limitations. Hot plate welding, though widely used, can lead to uneven heating and material degradation, particularly for temperature-sensitive polymers, as it requires direct contact with heated plates [27]. This can also result in issues like warping or dimensional changes in thin parts. Vibration welding, which uses high-frequency mechanical vibrations to generate heat, is effective for joining larger parts but can be challenging when working with thin or delicate materials, as the vibration forces may cause material distortion or cracking [28]. Induction welding, typically used for polymeric materials with conductive layers, faces challenges in controlling the depth of heat penetration and achieving consistent welds across varying thicknesses of non-conductive polymers [29]. Ultrasonic welding is effective for small components and thin materials but is less suitable for thick or large parts, where achieving a strong bond becomes difficult [30]. It also requires precise control of vibration amplitude, and improper settings can cause defects like cracking in rigid or brittle polymers. Microwave welding, while offering precise and localized heating, is not widely adopted due to its limited applicability to specific polymer types and the high cost of equipment [16]. Spin rotation welding, often used for thermoplastic materials, requires precise control over rotational speed and pressure to avoid material degradation or weak joints, and it may be less effective for joining parts with complex shapes or differing thicknesses [18]. FSW, which generates heat through mechanical friction between a rotating tool and the material, has shown promise for welding polymers, offering advantages such as low thermal degradation and the creation of strong, homogeneous bonds [31]. It allows for excellent control over the welding process and can be used for a variety of polymeric materials, especially those with higher viscosities [32]. However, FSW encounters challenges when welding thin and complex geometries, along with the need for precise tool design and optimization. [33], and it may not be suitable for polymers with low viscosity or those that are highly sensitive to temperature fluctuations [34]. Additionally, FSW equipment tends to be more complex and expensive, limiting its cost-effectiveness for small-scale production or applications requiring quick turnaround times [35].

In modern industry, the laser welding process (LWP) for polymeric materials has gained increasing interest because of its adaptability, versatility, minimal flash production, non-contact operation, ability to create deep and narrow welds, and the high-quality results it delivers [36]. In addition, LWP produces high-quality joints with high joint strength (JS) as well as small weld seam width (WSW). Moreover, it has good capabilities to weld products with complex geometries, accessibility to small and hidden areas that are difficult to weld by other means and maintain the alignment between the two components due to the non-contact nature of laser welding. LWP of thermoplastic materials has shown promising applications in different engineering fields including automotive (Fig. 2 shows examples of laser-welded automotive parts), aerospace, electronics, microfluidic devices, and medical devices [37,38]. For this industrial prevalence, researchers are making a great effort to improve the laser weldability of polymers [39]. The field of laser welding of thermoplastic materials has witnessed significant advancements and challenges, reflecting ongoing research and technological improvements. Understanding the current hotspots in this area helps to

highlight how recent developments align with or address these challenges. Fig. 3 shows the research hotspots regarding laser welding of thermoplastic materials [40–49]. The weldability of polymers is a key area of research, given the diverse thermoplastics used in industry [50]. Recent advancements have enhanced our understanding of how different polymers behave during laser welding, leading to better control over process parameters and improved weld quality. Researchers are now adept at addressing polymer-specific responses to laser heat, resulting in more consistent and reliable welding outcomes [51]. Compatibility issues between various thermoplastics and metals remain challenging, but progress in material science is addressing these problems by developing new alloys and composites that facilitate effective welding [52]. Innovations in welding techniques, such as multi-beam lasers, hybrid laser welding, and advanced focusing optics, are further advancing the field by improving precision and reducing defects, overcoming the limitations of traditional methods, and driving the development of more sophisticated welding solutions [40]. Recent research into laser welding has enhanced our understanding of the joining mechanisms, including how laser energy interacts with polymer surfaces to facilitate melting and bonding. Improved modeling and simulations are refining these processes, leading to better control and higher-quality joints [23]. Fracture modes in laser-welded joints, such as ductile or brittle fractures, are better understood thanks to advances in material modeling and testing, which help design more robust welding processes [38]. Achieving high weld strength remains a critical goal, with advancements in process control and optimization techniques, including computational methods and metaheuristic algorithms, enhancing weld quality [37]. Modeling and optimization through finite element analysis (FEA) and response surface methodology (RSM) are providing more accurate simulations and improving the efficiency of welding processes by identifying optimal conditions. Controlling weld seam width is vital for consistent quality, with advances in laser technology and process control systems allowing for more precise seam dimensions. Research into laser absorbers aims to improve energy absorption without additional coatings, while innovations in cooling techniques address thermal damage to polymers and surrounding areas by optimizing temperature control [53]. Welding dissimilar materials presents challenges like thermal mismatch, but recent advancements are developing techniques and materials to improve joint quality [54]. For similar polymers, research focuses on optimizing parameters to enhance weld quality. Accurate temperature control is crucial for uniform heating and cooling, with advancements in thermal modeling improving the ability to manage temperature profiles. Efforts to achieve cleaner, more visually appealing welds are supported by improved process control, while advancements in defect detection and process monitoring help prevent and address defects effectively [55]. Research is also enhancing the microstructure of weld joints to boost strength and durability [56,57]. Additionally, efforts to eliminate the need for laser absorbers are exploring alternative techniques, and advancements in achieving clear or transparent welds are focused on improving clarity and optical properties for specialized applications.

Generally speaking, the LWP of polymeric materials is more sensitive to the nonlinear temperature-dependent optical and thermophysical properties of welded materials, fabrication history, and additives. LWPs can be classified into LTW [23,37,58,59] and laser butt welding (LBW) [60–62] as represented schematically in Fig. 4. In LTW, one transparent polymeric sheet is put on another opaque sheet, and the laser beam is passed through the transparent sheet and absorbed by the opaque sheet [63–65]. The absorbed laser light is converted into heat which causes melting of the material at the weld interface to produce a lap welded joint. In LBW, the laser beam is projected on the interface of two adjacent polymeric sheets with a butt configuration [66–68]. The laser light is absorbed by the welded materials and is converted into heat, which causes the melting of the material at the weld interface to produce a butt welded joint. In laser welding of thermoplastic materials, several factors are affecting the quality of the welded joint as shown in Fig. 5. The

Table 1
Comparison between plastic welding processes from different perspectives.

Welding process	Requirement to filler materials/ implant (gasket)	Requirement to specific fixture/ backing	Limitation to part geometry	Cost to change from one product to another	Capability to weld long paths	Capability to weld metals	Dependence on optical properties of plastics	Existence of tool-workpiece interaction/ sticking	Clamping force	Formation of flash	Production rate	Repeatability	Initial cost	Popularity in the plastic welding industry
Laser welding	R/NR	NR	NL	NC	C	C	D	NE	L	L	H	H	H	P
Hot plate welding	NR	R	NL	HC	NC	NC	ID	E	H	H	M	L	M/H	P
Hot gas welding	R	NR	NL	NC	C	NC	ID	E	L	H	L	L	L	P
Extrusion welding	R	NR	NL	NC	C	NC	ID	E	L	H	M	M	M	P
Spin welding	NR	R	L	HC	NC	C	ID	NE	H	M	H	H	H	P
Vibration welding	NR	R	L	HC	NC	C	ID	NE	H	H	H	H	H	P
Ultrasonic welding	NR	R	L	HC	NC	C	ID	NE	H	M	H	H	H	P
Friction stir welding	NR	R	NL	NC	C	C	ID	E	H	M	L	M	H	NP
Induction welding	R	NR	NL	MC	NC	C	ID	NE	L	L	H	H	M	P
Microwave welding	R	NR	NL	NC	NC	C	ID	E	L	L	H	H	M	NP
Resistance welding	R	NR	L	MC	NC	C	ID	E	L	L	H	H	M	NP
Abbreviations	NR: Not Required R: Required		L: Limited NL: Not Limited NC: No Cost HC: High Cost MC: Medium Cost		C: Capable NC: Not capable		D: Dependent ID: Independent E: Exist NE: Not Exist		H: High M: Medium L: Low			P: Popular NP: Not popular		

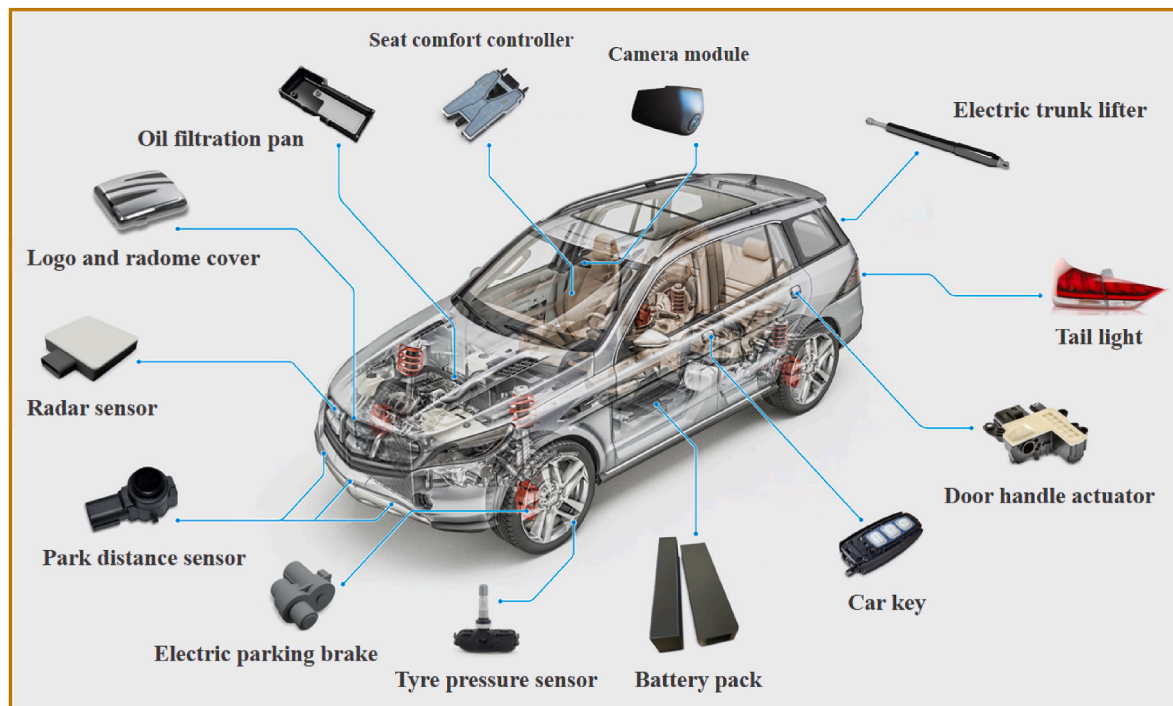


Fig. 2. Examples of laser-welded automotive parts (LPKF Laser & Electronics) [69].



Fig. 3. Research key topics in laser welding of polymeric materials.

influence of the majority of these factors on the process characteristics will be discussed in the next sections.

This paper provides a comprehensive review of the recent advances in laser welding technology as it applies to the joining of polymeric materials. The paper begins by introducing the various types of lasers employed in welding polymers, emphasizing the unique characteristics and applications of each. Initially, the focus will be on the CO₂ laser, an infrared laser with a wavelength well-suited for polymer absorption, known for its precision and efficiency in welding. The discussion will then move to the Diode laser, which is prized for its compact size, efficiency, and ability to perform high-speed welding, making it a popular

choice in the polymer industry. The review will also cover the Nd: YAG laser, highlighting its high power and versatility, which make it suitable for tasks requiring deep penetration and fine welding. Finally, the laser section concludes with the Fiber laser, which is celebrated for its excellent beam quality, energy efficiency, and flexibility in welding complex polymer shapes. The paper will then transition to discussing Laser absorbers, exploring the materials and techniques used to enhance the optical properties of polymers, thereby improving weld quality and broadening the range of materials suitable for laser welding. Following this, the review will delve into various LWPs, starting with similar welding, which involves the fusion of identical polymers, discussing process parameters and associated challenges. The focus will then shift to Dissimilar welding, addressing the complexities and strategies involved in joining different types of polymers. Next, Micro-welding will be examined, emphasizing the precision and control required for welding very small polymer components. This section will conclude by exploring the Welding of polymeric materials to metals, focusing on the unique challenges and innovative solutions associated with hybrid material joining. The paper will then explore the modeling of laser welding of polymers, discussing the importance of numerical, analytical, statistical, and artificial intelligence (AI) approaches in understanding the physical and thermal dynamics of the welding process to predict outcomes and optimize performance. In the final section, the review will discuss Advanced modeling techniques, including numerical simulations and machine learning, which are transforming the design and optimization of LWPs by offering deeper insights and improving predictive accuracy for industrial applications. The paper aims to provide a thorough understanding of the current state of laser welding technology and its potential to innovate and enhance the welding of polymeric materials.

2. Lasers used in the welding of polymeric materials

There are four main laser types used in the welding of polymeric materials, namely Nd: YAG, fiber, diode, and CO₂ [70]. The main characteristics of these types are represented in Fig. 6.

Laser power affects the amount of heat input, where higher power

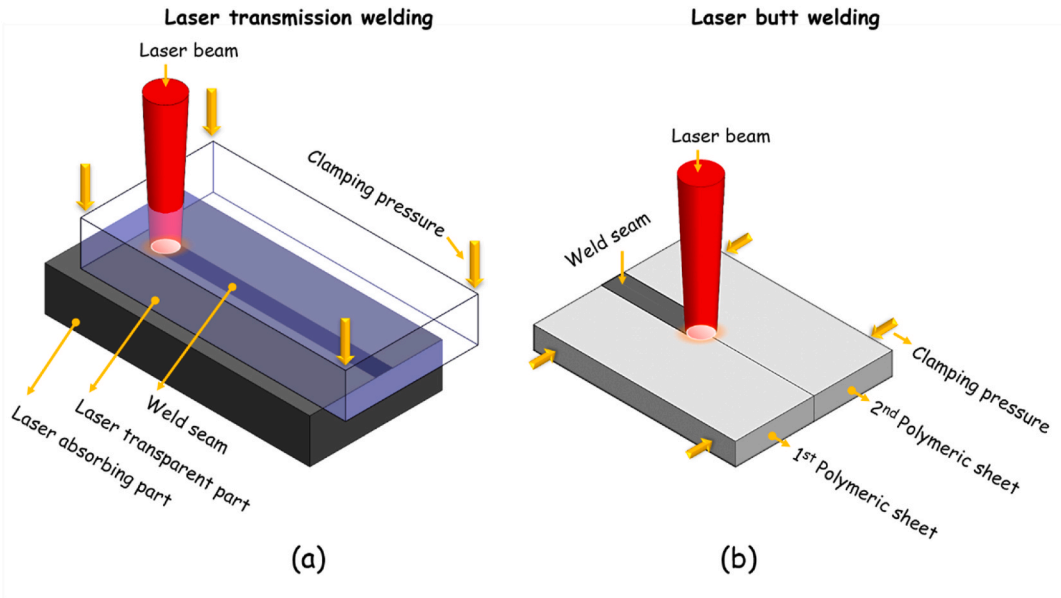


Fig. 4. Schematic of the LWP of polymeric materials: a) LTW and b) LBW.

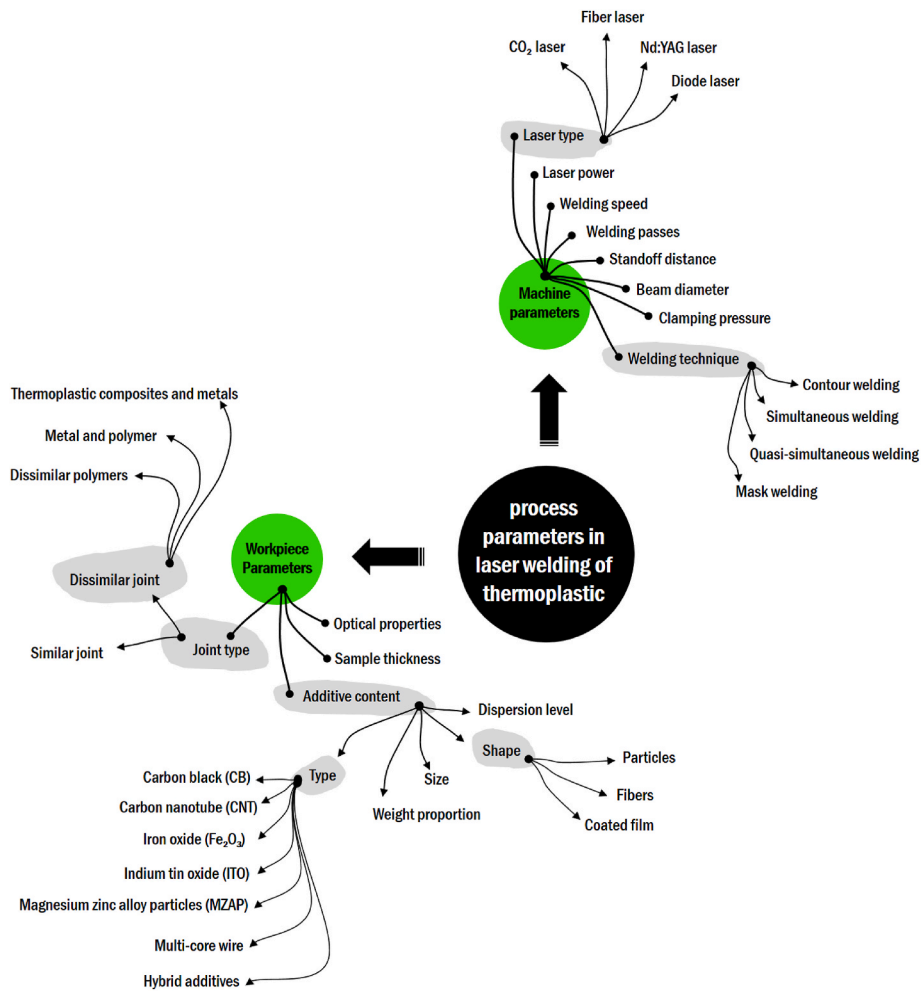


Fig. 5. Process parameters in laser welding of thermoplastic materials.

Laser system	CO ₂	Diode	ND:YAG	Fiber
Type	Gas laser	Semiconductor laser	Solid-state laser	Solid-state laser
Active medium	Mixture of CO ₂ , N ₂ and He	P-N junction diode (single crystal of gallium arsenide)	Yttrium Aluminium Garnet (Y ₃ Al ₅ O ₁₂)	Optical fiber doped with rare-earth elements
Delivery system	Mirror lens	Lens, optical fiber	Lens, optical fiber	Lens, optical fiber
Efficiency (%)	5-10	30-60	3-5	20-50
Beam quality	High	Low	High	High
Wavelength (nm)	10600	405-1940	1064	1000-2100

Fig. 6. Characteristics of lasers used for welding thermoplastic materials [53].

can enhance fusion but risks overheating, while lower power may lead to incomplete melting and weak joints [71]. Scan speed determines the duration of heat application, with higher speeds potentially causing insufficient fusion and lower speeds increasing the risk of thermal damage. The number of scans influences weld depth and uniformity; multiple scans can improve fusion but may cause overheating if not controlled properly. Laser diameter impacts heat distribution, with a larger diameter providing a broader but less precise weld area, and a smaller diameter offering better focus but requiring careful control to avoid overheating. Analyzing these parameters in detail helps in optimizing the welding process to achieve desired joint strength, thermal stability, and overall weld quality, making the review more comprehensive and actionable.

The interplay between laser parameters, such as beam scanning speed (BSS) and laser irradiance power (LIP), and polymer-specific thermal degradation mechanisms plays a crucial role in determining the quality and durability of welded joints in polymers like Polyamide 6 (PA6) and Polycarbonate (PC) [72]. Variations in BSS and LIP directly affect the thermal profile during welding: higher LIP increases the rate of heat generation, which can accelerate thermal degradation if not matched with an appropriate BSS. For PA6, a high LIP combined with a slow BSS may lead to excessive heat buildup, resulting in significant thermal degradation, such as chain scission and color change, which compromises the mechanical properties and long-term reliability of the weld. In PC, a high LIP could lead to rapid decomposition, including the release of volatile by-products that may cause defects such as porosity or weak bonding if the BSS does not adequately disperse the heat. Conversely, a faster BSS with moderate LIP can help mitigate degradation by reducing the exposure time and thus the extent of thermal degradation, but it must be carefully calibrated to avoid insufficient heating or incomplete melting. The degradation processes, including polymer chain breakdown and potential crosslinking or volatilization, critically impact the mechanical integrity of the weld, influencing tensile strength, impact resistance, and overall joint durability. Ensuring optimal laser parameters is essential to balance the energy input with the polymer's thermal stability, thereby enhancing the reliability and performance of the welded joints [73].

2.1. CO₂ laser

CO₂ laser has been employed to weld polymethyl methacrylate (PMMA), polypropylene (PP), polyethylene (PE), high-density polyethylene (HDPE), and low-density polyethylene (LDPE) [74–77]. However, it was applied only for low sheet thickness (less than 0.05 cm).

Moreover, it has a large size and large wavelength which prohibit the sharp focus of its radiation. Thus, the application of this type of laser in LWP of polymeric materials is very limited. CO₂ lasers offer high power, efficiency, and versatility, making them suitable for cutting, welding, and engraving a wide range of materials, including metals, plastics, and ceramics. They provide precision and accuracy, resulting in high-quality outputs, and are cost-effective for high-volume production with relatively low maintenance needs. However, they are often large and complex, which can lead to higher initial setup costs and space requirements. Their operational costs can also be significant due to gas expenses and maintenance. Additionally, CO₂ lasers operate at a specific wavelength (10.6 μ m), which limits their efficiency with certain materials and requires reflective optics for metal processing, adding to system complexity. They may also be less effective for very thin materials compared to other laser types.

2.2. Diode laser

Diode laser is commercially available with different radiation wavelengths (750–1500 nm) and operates in the near-infrared reflectography (NIRR). This wide range of wavelengths gives diode lasers a great advantage in welding different types of polymeric materials. Moreover, this technology has low fixed/operating costs. Different polymeric materials have been successfully welded using diode lasers such as PA6, HDPE, PP, PC, PMMA, PA6.6, PA12, PS, PBT, PC, dissimilar joints made of glass-fiber-reinforced polyamide (GF-PA) and glass-fiber-reinforced polypropylene (GF-PP), dissimilar joints made of glass-fiber-reinforced acrylonitrile butadiene styrene (GF-ABS) and glass-fiber-reinforced polypropylene (GF-PP), dissimilar joint made of glass-fiber-reinforced polycaprolactam (GF-PA6) and PC [50,78–81]. However, this type of laser has severe design difficulties in order to produce a sharp beam focus. The large size of LS as well as its wide radiation spectrum may impair beam quality, especially for long operating distances [82]. Diode lasers offer several advantages, including compact size, high efficiency, and lower operational costs compared to other laser types. They are well-suited for applications requiring precise, high-quality outputs, such as in medical procedures and industrial tasks like cutting and welding of metals and plastics. Their solid-state design allows for easy integration into various systems, and they generally require less maintenance. However, diode lasers have limitations, such as a narrower wavelength range compared to CO₂ lasers, which can restrict their versatility with certain materials. They also typically have lower power outputs than CO₂ lasers, which can limit their effectiveness for high-volume or thick material processing. Additionally, their

performance can be influenced by temperature variations, requiring careful thermal management.

2.3. Nd: YAG laser

Nd: YAG laser operates in the NIR with a wavelength of 1064 nm and has been also used in welding different types of polymeric materials such as PE, PC, ABS, PA66, GF-PA66, PA6, C-PA [83–86]. The Nd: YAG laser has excellent beam quality and a small LS. However, this type of laser suffers from low absorptivity with different polymeric materials, which requires the addition of absorbing materials between the two polymeric sheets [83]. It offers several notable advantages, including high beam quality, the ability to operate at various wavelengths (1064, 532, 355, and 266 nm), and high power output, which makes them suitable for a wide range of applications such as cutting, welding, and medical procedures. They also feature good efficiency and durability, with relatively low maintenance needs. However, they have limitations such as higher operational costs compared to some other lasers and require complex optical systems to produce their shorter wavelengths. Additionally, they are less efficient for certain materials compared to CO₂ lasers, particularly when processing non-metallic materials. Their setup and alignment can also be more intricate, requiring precise calibration for optimal performance.

2.4. Fiber laser

Fiber laser operates with a wide range of wavelengths (1000–2100 nm) and can be readily absorbed by different types of polymeric materials such as ABS, PP, PA6, HDPE, PETG, POM, PLLA, PA, PMMA, PVC, PEEK and PC [43,87–92], without application of any absorbing additives. Fiber lasers exhibited superior quality and higher efficiency compared with other lasers [37]. Due to its high quality and efficiency, it could be used in the welding of thin films and micro-welding applications [82]. Fiber lasers offer numerous advantages, including high efficiency, excellent beam quality, and the ability to produce high-power outputs with compact and rugged designs. Their narrow beam diameter allows for precise cutting and welding, and their long lifespans and low maintenance requirements make them cost-effective over time. Additionally, fiber lasers are versatile, with the capability to process a wide range of materials, including metals, plastics, and ceramics. However, they can be expensive to purchase initially and require careful handling due to their sensitivity to dust and contamination. Fiber lasers also have limited performance when dealing with highly reflective materials, which can cause damage to the laser optics and reduce.

3. Laser absorber

Most polymeric materials are transparent in the spectral range of most lasers (800–1100 nm) [93]. In the past, LWP of polymeric materials was accomplished with CO₂ lasers which have a large wavelength as polymeric materials have excellent absorption capabilities for this wavelength domain [94–96]. However, the use of these high wavelength lasers in LWP of polymeric materials may have low penetration in most materials and cause undesirable damage to the weld surface [97]. Thus, lasers with short wavelengths were recommended to weld this kind of material. On the other hand, lasers with short wavelengths have a poor absorption coefficient in most transparent polymeric materials, and consequently, they cannot be directly used to heat, melt, or weld transparent polymers [98].

Laser absorbers play distinct roles in enhancing the optical properties of polymeric sheets for laser welding, with their impact varying between LTW and LBW. In LTW, where laser energy is transmitted through a transparent polymer layer to a second layer that absorbs the energy, the choice of absorber is critical for ensuring effective energy absorption and heat generation. Absorbers like carbon black (CB) and metal oxides are used to optimize the energy transfer through the transparent layer, with

CB providing broad-spectrum absorption and metal oxides allowing for precise wavelength tuning, thus improving weld quality and reducing the risk of weak joints. In contrast, LBW involves direct absorption of laser energy by the polymer surface, where absorbers enhance surface absorption to improve weld strength and appearance. Here, absorbers such as CB or metal powders increase energy absorption at the surface level, affecting factors like penetration depth and thermal profile. While both LTW and LBW benefit from effective absorbers, LTW requires materials that ensure efficient energy transfer through layers, whereas LBW focuses on optimizing surface interaction for better weld quality. LWP of transparent polymeric materials using conventional fiber, Nd: YAG or diode lasers requires applying an opaque thin film between welded materials which acts as a laser absorber [99]. The most commonly used deposited materials are dye, Ti, CB, or graphite [100–102]. Adding nanomaterials to polymeric sheets to enhance their optical properties for LWP has been extensively reported in the literature [61,103,104]. Table 2 summarizes some of the materials used as absorbers in laser welding of polymeric materials. Visco et al. [105] welded a pristine PE sheet to another PE sheet reinforced with carbon nanotubes (CNTs) using Nd: Yag laser. Different concentrations of CNTs (varies from 0.1 to 10 wt%) were employed to investigate their effects on the joint quality. Experimental results confirmed that the CNT filler helps in absorbing laser light effectively. The best CNT concentration was about 0.1 wt% which produced a sound joint with a tensile strength of 115 N. Nakhaei et al. [106] reinforced PP with nano-clay at different concentrations (0, 3, and 5 wt%) to enhance the absorptivity of the butt-welded sheets to CO₂ laser light. The results showed that an increase in BSS and nano-clay concentration impaired the weld strength while increasing LIP enhanced the weld strength. Additionally, JS increased with increasing focal position to moderate values (up to 1 mm) and then the JS started to decline. Hadriche et al. [107] added CB particles with 0.2 wt% to PP in the LTW process. Bates et al. [72] modeled the LTW of PA6 and PC reinforced by CB with different concentrations to predict the occurred degradation of the welded materials under time-dependent thermal loading. It was reported that, for low CB concentrations, degradation has occurred under higher LIP. This was attributed to the bigger material volume on which the laser light is absorbed and consequently, only high LIP may cause high temperatures at which the thermal degradation of the welded materials may occur. Acherjee et al. [108] added CB particles with 0.1 wt% to PC sheets which acted as an opaque sheet in LTW with a pure PC sheet. PMMA has been also reinforced with CB with 0.2 wt% to enhance its absorptivity to laser light during LTW to a pure PMMA sheet [109]. The effects of different welding parameters including LIP (19, 22, 25, and 28 W), BSS (240, 300, 360, and 420 mm/min), and standoff distance (28, 31, 34, and 37 mm) on the weld strength and WSW were investigated. It was recommended to use high LIP, low BSS, and moderate standoff distance to obtain sound joints with high strength. On the contrary, low LIP, high BSS, and low standoff distance were recommended to obtain a narrow weld seam. In another study, PC containing 0.2 wt% CB particles was also welded using the LTW technique [110]. Based on the results, a moderate LIP was recommended in order to obtain high-strength joints. CB was also added to ABS sheets at a concentration of 0.1 wt% to enhance its weldability to PC sheets during LTW [111,112]. CB with a concentration of 0.5 wt% was also added to PP sheets with different clay contents during LWP of a butt-joint configuration [113]. The effects of different welding parameters including LIP (40, 50, and 60 W), BSS (30, 45, and 60 mm/min), clay concentrations (0, 3, and 6 wt%), and standoff distance (0, 1 and 2 mm) on the weld strength were investigated. It was reported that the weld strength was significantly affected by the clay concentration followed by LIP, BSS, and focal position. To obtain high joint strength, it was recommended to use a LIP of 53 W and low BSS. Aden et al. [114] employed TiO₂ and CB particles as laser absorbers during LWP of PP which reduced the amount of power required to accomplish the welding process. However, the use of absorbers increases the cost of LWP due to the cost of absorbers as well as the preparation cost. Thus, developing

Table 2

Materials used as a laser absorber in plastic welding.

Laser absorber	Base material	Additive concentration	Laser type	Wavelength (nm)	Ref.
Carbon black (CB)	Isotactic PP	0.5, 1.0, 2.0 and 3.0 wt%	Diode	980	[117]
	PP	0 and 2%	Diode	1530	[114]
	PP	0.1 and 0.2 wt %.	Diode	980	[118]
	PET		Diode	808	[99]
	PC to ABS	0.1 wt %	Diode	809.40	[112]
Zinc particles mixed with 1% carbon black (ZPCB)	PA6, PA6GF and PC	Varies between 0 and 0.1 wt%	Diode	940	[80]
	PASF	—	Diode	980	[119]
	PETG	—	Fiber laser	1064	[120]
Carbon nanotube (CNT)	ABS	0.01% and 0.05% wt. %	Diode	808	[121]
	PE to PE-CNT	1 wt %	Nd:YAG	532	[122]
Silver nanoparticles (AgNPs)	ETFE	—	Diode	808	[123]
Indium tin oxide (ITO)	PP	0 and 2%	Diode	1530	[114]
Multi-core Cu wire	PMMA	—	Diode	915	[124]
Iron oxide (Fe ₂ O ₃)	UHMWPE	Varies between 0.1 and 10 wt%	Nd:YAG	532	[125]

advanced lasers to accomplish LWP without any absorber is highly desirable. The wavelength of these tailored lasers is adapted to match the absorption bands of the welded material. For instance, the adaptation of wavelengths is achieved in fiber lasers by doping the fibers with active elements such as Th (1950 ± 150 nm), Er (1560 ± 40 nm), or Yb (1070 ± 10 nm). Roesner et al. [115] investigated absorber-free LWP of different polymeric materials such as PP, POM, PMMA, PVA, and PBT. Mamuschkin et al. [116] developed a pulsed irradiation technique to achieve absorber-free LWP of polymeric materials. They employed laser sources with wavelengths of 1750 ± 250 nm to weld polymeric materials by exploiting intrinsic absorption characteristics of thermoplastics without any light absorbers. The laser light is absorbed along the propagation direction according to the Lambert-Beer law. Roth et al. [68] investigated the LWP of cyclic olefin copolymer (COC) by infrared femtosecond laser without applying an absorption layer. To obtain WSW of 38–137 μ m, the employed laser had LIP of 500–1700 mW, a wavelength of 1028 nm, a repetition rate of 57 kHz and a pulse duration of 220 fs. A high BSS of 20 mm/s was achieved in single-pass welding.

Nanotechnology offers transformative potential in optimizing the laser welding of polymeric materials by modifying surface properties to enhance both laser absorption and heat dissipation. By incorporating nanomaterials such as metallic nanoparticles, carbon nanotubes, or graphene into the polymer's surface or using them as coatings, the polymer's optical properties can be precisely tuned to increase laser energy absorption. These nanomaterials can be engineered to absorb specific wavelengths of laser light, improving the efficiency of energy transfer into the weld zone. This targeted absorption leads to stronger and more consistent welds. Additionally, nanotechnology can enhance heat dissipation by integrating nanoscale fillers that improve the polymer's thermal conductivity. Materials like graphene or boron nitride can create efficient pathways for heat transfer, reducing localized overheating and minimizing thermal damage or warping. This improved thermal management allows for higher welding speeds and more precise control of the welding process. Furthermore, self-healing or adaptive nanocoatings can be designed to respond dynamically to changes in temperature, helping to manage thermal gradients and repair minor defects during the welding process. Such innovations in nanotechnology not only enhance the performance and durability of the welds but also increase the overall efficiency of the welding operation, opening new avenues for advanced polymeric material applications.

4. Laser welding processes

In this section, the main LWPs will be discussed including similar welding, dissimilar welding, micro-welding, and welding polymeric materials to metals.

4.1. Similar welding of polymeric components

In similar welding, two polymeric components made of the same materials are welded together using LTW or LBW techniques. In similar laser welding of thermoplastic materials, critical factors influencing joint quality include the precise control of laser power, focus, and speed, as well as the compatibility of the materials' thermal properties and absorption characteristics. The choice of laser type, whether it is a continuous-wave or pulsed laser, directly impacts the heat input and penetration depth. For instance, continuous-wave lasers are effective for steady and uniform heating, while pulsed lasers can be used for finer control over heat application, reducing the risk of thermal degradation. The interplay between these factors ensures consistent weld quality by optimizing energy distribution and minimizing defects such as incomplete fusion or warping. Light absorbers may be used to enhance the absorptivity of the welded materials. Fernandes et al. [126] investigated the weldability of the high-performance thermoplastic, white ERTALON® 6 SA, using a pulsed Nd:YAG laser. Tensile tests evaluated joint quality and strength, supported by a numerical model to validate theoretical findings. The research addressed the challenges of welding two transmitting components, especially white thermoplastics, which are more difficult to weld than transparent parts. A clamp system ensured contact between samples, eliminating gaps and improving heat conductivity. A parametric study tested various Nd:YAG laser settings, enhancing weld quality and achieving optimal results at a laser energy density of 8.26×10^3 J/cm². However, tensile tests showed the joint was weaker than the base material. Future research should consider higher energy densities, beam sizes, and overlapping to improve the process. The numerical model identified joint weaknesses, simulating tensile tests to detect local stress concentrations, especially at joint ends, which caused premature failure under low loads. The joint's geometry contributed to shear during loading. Despite joint strength being lower than the base material, this research advances the understanding of welding white polymers, which are challenging to weld, by identifying failure mechanisms and optimizing parameters for improved strength. Notably, this is the first study to weld Ertalon 6 SA white components using a pulsed Nd:YAG laser. Hadriche et al. [107] used LTW to weld PP sheets. The laser energy was absorbed by the PP sheet and was converted into heat which resulted in rising the interface temperature to reach crystallization conditions that are lower than the melting temperature and the glass-transition temperature. It was reported that the crystallinity depended on cooling and heating rates that affected the crystallization kinetics related to recrystallization and melting processes. HDPE sheets were successfully welded using a high-power diode laser [127]. Different BSS of 1–3 mm/s and LIP of 200–460 W were used during welding. The weld quality of LWP outperformed the hot-tool joining technique. Goyal and Kant [128] explored the feasibility of underwater LTW of PC sheets by examining the impact of water depth, laser power, beam diameter, and scan speed on weld strength, dimensions, and bond

morphology, comparing these factors with air welding conditions. The experiments showed that underwater conditions could improve weld quality, with a maximum breaking force of 567.5 N at a 20 mm water depth, which is 6% higher than in air, attributed to reduced thermal degradation and a broader bubble-free region. The breaking force increased with laser power, scan speed, and beam diameter up to a point, before decreasing. The maximum was achieved at 110 W, 200 mm/min, and a 6 mm beam diameter. Weld width increased with laser power and beam diameter but decreased with scan speed, with greater width observed in the air due to less scattering and absorption of laser energy. As water depth increased from 0 to 30 mm, the interface temperature dropped from 1063 to 490 °C, with breaking force initially rising due to enhanced polymer chain interdiffusion, before decreasing due to thermal degradation and burning. Cross-section micrographs indicated that molten depth decreased with rising water levels, with greater depth in the upper transparent part due to volumetric absorption. Optical microscopy (OM) and scanning electron microscopy (SEM) analyses showed that tunnels formed by large bubbles reduced weld strength, but a wider bubble-free melt region underwater contributed to better weld integrity. Juhl et al. [129] evaluated the quality of PS/PS lap joints welded by diode laser (808 nm). The effects of repetition time on interpenetration depth at the welding interface and the mechanical properties of the welded joint were investigated. They also explored the fracture mechanisms in the welded joint interface; namely chain rupture and chain pull-out. Weld strength varied from 20.27 to 21.56 kJ/m² for a wide range of BSS (105.3–500.0 mm/s). The weld strength is approximately constant for all BSS and line energy, and full strength is obtained at BSS of 500 mm/s and line energy of 100 J/m. The investigated welding conditions resulted in WSW varying from 1.50 to 1.88 mm and it is inversely proportional to BSS. LTW was also employed to join PC sheets and the RSM was employed to design the experiments considering central composite rotatable methodology [130]. The established mathematical models obtained by RSM were evaluated using analysis of variance (ANOVA). Process control variables were the LIP, BSS, clamping pressure, and scanning number; while the process responses were JS and WSW. It was reported that increasing LIP increases the JS and WSW, while both the responses decreased by increasing BSS. The CP enhanced the joint quality as it guarantees excellent contact between the welded sheets which consequently enhanced the heat transfer process by conduction mode via the contact surface. However, CP had an insignificant effect on the investigated process responses compared with that of LIP, BSS and scanning numbers. In another study, Acherjee et al. [108] welded PC sheets using LTW technique. They investigated the effects of LIP, BSS, clamping pressure and standoff distance on the JS and WSW-based statistical sensitivity analysis. It was reported that the JS is more sensitive to BSS followed by LIP than other welding parameters. They had a sensitivity of 84 and 30.8 N/cm per unit BSS and LIP, respectively. Moreover, WSW is more sensitive to BSS followed by standoff distance than other welding parameters. They had a sensitivity of 0.37 and 0.11 mm per unit BSS and standoff distance, respectively. RSM was also employed to design the experimental plan during welding two PMMA sheets by LTW technique using low power Nd:YVO₄ laser [131]. The effects of welding parameters including LIP, BSS and pulse frequency on the joint quality were investigated. Sensitivity analysis was carried out to explore the effectiveness of the process parameters on the joint quality in terms of JS and WSW. LIP was declared as the most sensitive parameter for both quality measures and BSS was the main factor affecting both quality measures.

In addition to joint strength and weld seam width, evaluating the performance of laser-welded polymeric materials requires a comprehensive assessment of several other critical metrics. Thermal stability is a crucial factor, as the welds must withstand operational temperatures without degrading or losing their mechanical properties. Assessing the surface finish of the weld is also important, as the quality of the weld surface can impact both aesthetics and functionality, particularly in applications requiring high visual standards or smooth interfaces. Long-

term durability is another key metric, which involves evaluating how well the welds perform over extended periods under various environmental conditions such as ultraviolet (UV) exposure, moisture, and thermal cycling. Microstructural integrity, including the presence of voids, cracks, or delaminations within the weld, can significantly affect the overall strength and longevity of the weld. Chemical resistance is vital for applications where the welded components are exposed to aggressive substances, ensuring that the welds do not degrade or fail when in contact with chemicals. Furthermore, process consistency and repeatability are essential for ensuring uniform weld quality across multiple units in production settings. Evaluating these additional characteristics provides a more holistic view of weld quality and performance, enabling better predictions of how the welded polymeric materials will behave in their intended applications.

4.2. Dissimilar welding of polymeric components

In dissimilar welding, two components made of two different materials are welded together using LTW or LBW. Welding two dissimilar polymeric sheets is a cumbersome industrial problem due to the large disparity in properties between different polymeric materials. In LTW of dissimilar joints, a transparent component is put on the opaque component. The laser passes through the transparent component and is absorbed by the opaque component. The absorbed energy is transformed into heat, which is used to melt the welded components at the contact interface [132–134]. When welding dissimilar thermoplastic materials, factors such as the difference in melting points, thermal expansion coefficients, and absorption properties become crucial. The choice of laser type needs to accommodate the distinct thermal characteristics of each material, often requiring precise adjustments in laser power and beam modulation. For example, a pulsed laser might be preferred to manage the different melting behaviors and prevent overheating. The selection of appropriate absorbers and their distribution also plays a role in ensuring that both materials are adequately heated and fused, impacting the overall joint integrity and performance. Achieving hermetic seals in laser-welded automotive components made of dissimilar polymers like PBT and PC is increasingly feasible due to advancements in laser technology, though several critical challenges must be addressed. Technological improvements such as high-precision lasers and advanced beam delivery systems have enhanced the ability to weld dissimilar polymers with high accuracy, enabling effective hermetic sealing. However, minimizing welding displacements and optimizing laser beam angles are crucial to achieving consistent weld quality, as any misalignment can lead to leaks or weak joints. Managing thermal stresses is another challenge, as differential thermal expansion and contraction between PBT and PC can induce warping or internal stresses, compromising the structural integrity of the seal. Compared to traditional welding methods, such as ultrasonic or vibration welding, laser welding offers superior precision and control, allowing for smaller weld spots and less thermal impact on adjacent areas. This precision is critical for creating high-integrity seals in complex automotive components. However, traditional methods may still be preferred in certain contexts for their robustness and simpler process setup, particularly in high-volume production environments. Laser welding's unique advantages include its ability to produce cleaner, more precise welds with reduced post-processing requirements, but it may come with higher initial costs and complexity. Balancing these factors is key to optimizing the welding process for automotive applications. Acherjee et al. [111] investigated the LTW of ABS and PC sheets. They investigated the effects of LIP (8, 11, 14, 17, and 20 W), BSS (4, 7, 10, 13, and 16 mm/s), CP (0.9, 1.8, 2.7, 3.6, and 4.5 MPa), and stand-off distance (25, 30, 35, 40 and 45 mm) on the track width and JS. LIP was reported as the most significant factor affecting the JS followed by focal distance and BSS. CP was reported as an insignificant factor while the focal distance was reported as the most significant factor affecting the track width followed by LIP, BSS, and clamping pressure. In another study, Shin and Choi [135] investigated

the effects of different welding parameters on the joint quality during welding of ABS to PC using LTW. HAZ and JS were measured for different experimental conditions. At the low BSS, the welded surface contained carbonization damage resulting from the high heat input. Goyal and Kant [136] investigated LTW of PS and PC using electrolytic FE powder as an absorber, focusing on how laser power and scan speed affect weld strength and bond morphology. Lap shear tests evaluated breaking force and ductility while SEM and OM examined the fractured surfaces and cross-sections. Results showed that laser power and scan speed significantly impacted weld strength and ductility. Micrographs indicated that FE particle anchoring, polymer interdiffusion, and small bubbles enhanced bond strength, whereas thermal degradation and large bubbles weakened it. The study found that weld strength and width were influenced by line energy, with a maximum strength of 4.12 MPa at 150 W laser power and 11 mm/s scan speed. Brittle fractures were observed due to polystyrene's nature, but higher laser power increased ductility. Small bubbles improved polymer chain intermixing while burning and carbonization weakened joints. Cross-section analysis showed different zones in the weld seam due to energy variation, with thermal degradation likely at the center. Polymeric bridges in the heat-affected zone provided shear resistance, and FE particles at the interface enhanced bond strength through adhesive bonding. Bin In et al. [137] utilized a vertically aligned CNT structure produced by chemical vapor deposition technique in welding PC to PC or PMMA sheets using LTW technique as shown in Fig. 7. The developed CNT structure had excellent absorption characteristics which produce selective heating at the interface with polymeric materials. Moreover, its compliant and porous nature averts air pockets formation at the welding interface. Low LIP produced weak adhesion; while high LIP produced periodic damage in the welding interface. Mingareev et al. [90] investigated the LWP process of different polymeric materials (LDPE, HDPE, POM, PP, PETG, and PMMA) using a continuous laser with a wavelength of 2 μm to produce laser-welded butt joints. Moreover, they investigated the LTW of PMMA, PETG, and combinations of PP/PMMA and LDPE/PP. The JS of LDPE and HDPE exceeded 80% compared to base material strength. Many welded materials had better absorptivity at the used wavelength compared with that at wavelengths around 1 μm . Accomplishing LWP at that wavelength gave considerable importance to that technique as an eye-safe and low-energy welding technique for polymeric materials. Mehrpouya et al. [138] investigated LTW of pure PET

and PET-PEVA blends with concentrations of 5, 10, and 15% using a diode laser. They investigated the effects of LIP and BSS on joint elongation and maximum load. PET-PEVA blends had a close thermal characteristic to pure PET and the use of PEVA enhanced the mechanical properties of the welded joints in terms of both elongation and maximum load. PET-PEVA 15% blend exhibited better weldability to pure PET compared to the other investigated blends. Lakemeyer and Schöppner [139] welded real automotive components made of PBT and PC using LTW without applying any CP. It was reported that increasing LIP may reduce cycle time with a minimum number of scans. In addition, larger welding displacements were observed in the middle of the welded component due to the small angle between the material and the laser beam. Nevertheless, to get a hermetic seal around the welded component, very high welding displacements should be applied due to the warping effect of the injection-molded components. Thermal degradation that occurs in welded joints made of PA6 and PC during LTW was investigated in Ref. [72]. They observed that the degradation of the polymers may cause a decrease in strength. Generally speaking, polymer degradation is a function of both temperature and time. This degradation is attributed to low BSS and/or high LIP laser power which result in high welding temperatures. Liu et al. [140] applied the grafting technique to improve the weldability of PA66 and PE using LTW as these two polymeric materials exhibited poor compatibility and significant polar differences. It was reported that the thermal and mechanical properties of PE were not significantly changed after grafting modifications. Nevertheless, a high JS was obtained in the case of modified PE. Micro-morphology of the welded region indicated that grafting modification caused sturdy interfacial adhesion and good compatibility between welded materials. Beam wobbling effects on LTW of welded joints made of dissimilar polymeric materials were investigated in Ref. [141]. PMMA and PC were efficiently welded using wobble LTW applying a low-power Nd:YVO₄ laser. Beam wobbling had a considerable effect on the shear strength of the welded joints. The wobble width had the supreme effect on the shear strength of the welded joints, followed by BSS, wobble frequency, LIP and pulse frequency. The beam wobbling resulted in homogenized heat input, good heat transfer, turbulence flow in the melted pool, and good mixing between welded polymers. These effects are the main reasons for augmenting weld quality.

Dissimilar polymer welding presents significant challenges due to the inherent differences in their thermal, chemical, and mechanical

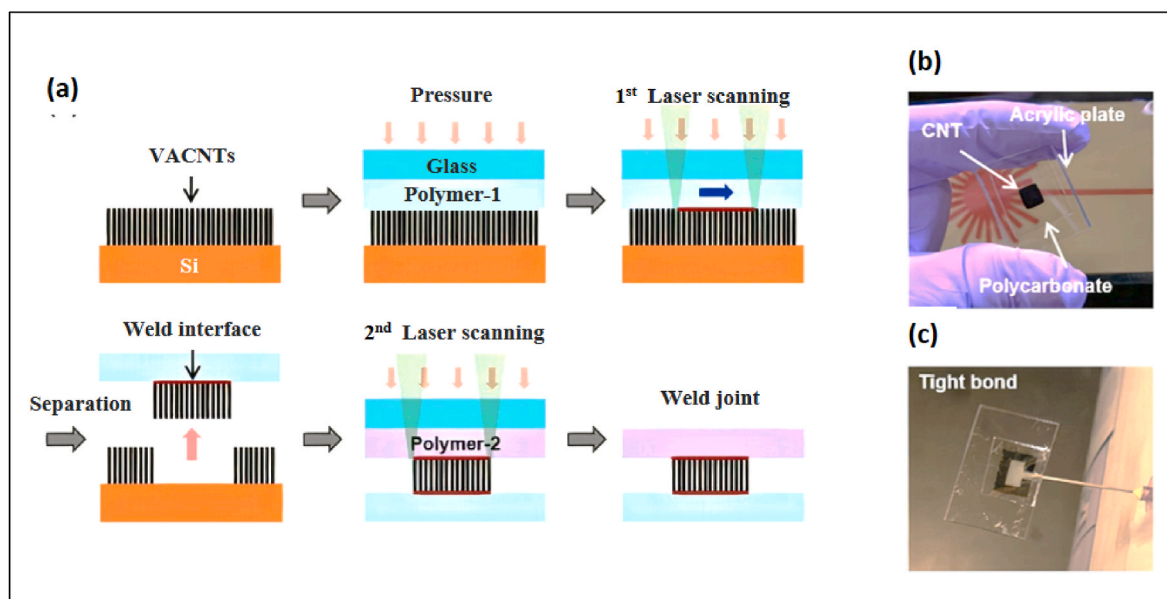


Fig. 7. (a) Schematic of the LTW technique, (b) Photograph of PC-PMMA joint with vertically aligned CNTs structure, and (c) Photograph of PC-PC joint with vertically aligned CN structure [137].

properties. These disparities can result in issues such as mismatched melting points, which may cause one material to degrade while the other is not sufficiently melted, leading to weak or incomplete welds. Additionally, differences in thermal expansion coefficients can cause differential expansion and contraction during the welding process, creating residual stresses and potential warping or cracking. Chemical incompatibility between the polymers can hinder intermolecular bonding, leading to poor adhesion and weak joints. Mechanical property differences, such as varying stiffness and impact resistance, can result in stress concentrations at the weld joint, increasing the risk of failure under load. To address these challenges, several strategies are employed, including precise control of welding parameters to optimize temperatures and heating rates, the use of interlayers or tie layers to improve bonding between the polymers, and the application of compatibilizers or polymer blends to bridge chemical differences. Innovative welding techniques such as sequential welding, advanced laser beam shaping, and surface treatments like plasma activation can also enhance compatibility and joint quality. Post-welding treatments, including annealing and protective coatings, further help to improve the strength and durability of the welds, ensuring the reliable performance of the final product.

4.3. Micro-welding

Micro-welding in the laser welding of polymers is critically important due to its ability to enable precise, high-quality bonding in applications requiring intricate and miniaturized components, such as in the electronics, medical devices, and microfluidics industries. This technique facilitates the joining of small parts with sub-millimeter dimensions, ensuring robust and reliable connections without compromising the delicate structures of the polymer components. The precision of laser micro-welding allows for minimal thermal impact on surrounding areas, which is essential for preserving the integrity and functionality of sensitive polymeric materials. Additionally, the ability to control the laser energy with high accuracy results in strong welds with fine resolution, making it possible to create complex geometries and intricate patterns that are critical for advanced applications. As industries continue to demand increasingly smaller and more sophisticated polymer-based devices, micro-welding becomes indispensable for meeting these stringent manufacturing requirements and achieving high-performance, durable products. In the micro-welding of thermoplastic materials, the critical factors include the fine control of laser parameters, such as very low power settings, precise focus, and rapid scanning speeds. The laser type, typically a high-precision fiber or UV laser, must be capable of achieving the sub-micron resolution required for micro-welding [142]. The ability to finely tune these parameters ensures minimal thermal impact on surrounding areas and accurate joint formation. Effective energy management and precise heat application are essential to achieve strong, defect-free welds in small-scale applications where even minor deviations can significantly impact quality.

Achieving sub-micron resolution in micro-welding of polymers is constrained by the interplay between laser power, wavelength, material properties, and laser system capabilities. Theoretically, to achieve sub-micron resolution, lasers must focus on extremely small spot sizes, which are limited by the diffraction of light, governed by the wavelength of the laser. Shorter wavelengths, such as those in the UV range, offer the potential for finer resolution due to their reduced diffraction limits, enabling more precise control over the welding process. However, the chosen wavelength must align with the polymer's absorption characteristics; polymers have specific absorption bands, and lasers must match these to be effective. In terms of power, while higher laser power is needed to achieve localized melting and bonding at the sub-micron scale, excessive power can cause thermal degradation, excessive heat diffusion, or even vaporization of the polymer, compromising the resolution and weld quality. This necessitates a delicate balance and often the use of pulsed lasers with high peak powers in brief bursts to minimize heat spread and damage. The limitations are further influenced by

the polymer's thermal conductivity and melting point, which dictate how the laser energy is absorbed and distributed. Additionally, achieving sub-micron resolution demands advanced laser systems with high-precision optical components capable of focusing the beam to extremely small diameters while maintaining stability and delivering consistent energy. Innovations in laser technology, such as adaptive optics and high-resolution focusing optics, are crucial to overcoming these theoretical limits and achieving the desired precision in the micro-welding of polymers. Welding of micro parts such as microfluidic devices using LWP has experienced a considerable spread in the last few years [143–146]. The main challenge of this technique is to obtain a small LS size that is appropriate for micro-welding. Ussing et al. [147] successfully developed a focused low-power laser with an LS area of $50 \times 5 \mu\text{m}$ to obtain very small welding seams on PMMA micro-specimens. It was reported that the welding joint with a WSW of 25–40 μm had lower strength than that of the joint with a WSW of 300–500 μm . However, CP had a much more significant effect on the JS than WSW. Neuber and Heilmann [123] applied a thin film of Ag deposited by magnetron sputtering on 200 μm thick ETFE foil to enhance its absorptivity during LWP. The used laser was a continuous diode type with a wavelength of 808 nm. The applied film enhanced the absorption of laser energy during the irradiation process thanks to the plasmon resonance that occurs in nanoparticles. During the welding process, Ag flocs appear instead of the continuous Ag film. These newly formed flocs prevent the plasmon resonance process. The tensile strength of the welded joint was 30 N/mm² which is sufficient for membrane applications. Singh et al. [148] welded a thermoplastic biochip made of PMMA (VM100, V825T 101, and Plexiglass®) and COC (5013 and 8007) using LWP. A thin carbon film (1–10 nm thickness) was deposited onto the welded materials using a carbon coater. Nd: YAG laser with a wavelength of 1064 nm was employed during the welding process. The welded biochip had a thickness of $\sim 670 \mu\text{m}$ with a channel width of 50 μm . PMMA-Plexiglass® and COC-8007 had the best weldability compared with other investigated materials. Gisario et al. [149] welded 8006 Al film (50, 100, and 150 μm) coated with a PL resin (80 μm) with PP films (1100 μm) using a diode laser source with a maximum LIP of 1500 W and an elliptical focal spot ($1200 \times 3800 \text{ mm}$). The Al film was subjected to H22 heat treatment and an automatic film applicator was employed in the coating process. It was recommended to use a LIP of 120W and BSS of 7 mm/s in order to obtain a sound joint. Visco et al. [150] used an Nd: YAG laser with a wavelength of 532 nm, pulse energy of 150 mJ, pulse duration of 3 ns, and repetition rate of 10 Hz to weld PE thin film. Both pure PE film and reinforced PE film with CNTs were tested. The reinforced films absorbed laser energy more effectively than pure films. It was recommended to use low irradiation times to reduce erosion that occurred in the welded region.

4.4. Welding of polymeric materials to metals

Polymeric materials, including high-performance polymer-based composite materials, have low wear resistance and impact toughness. Therefore, polymeric materials are welded to metals to obtain complex structures that fulfill the engineering requirements of some aviation or automotive applications [151–155]. A brilliant property of that type of welded structure is the absence of interface corrosion [156]. Moreover, welding polymeric materials to metals can lighten the weight of the whole structure to lessen the consumption of energy [157,158]. In 2008, Katayama and Kawahito [159] developed a novel rapid LWP of polymer and metal sheets as shown in Fig. 8. They used a diode laser with a wavelength of 807 nm under a cooling nitrogen gas flowing at 35 L/min to weld a PET sheet to a 304-steel sheet. The PET/steel sheets absorb the laser energy and convert it into heat and bubbles are generated and expanded rapidly, and consequently, chemical and/or physical bonding occurs at the interface between the two sheets. PET was also welded to 1060 Al allowed using the LWP technique [160]. To produce high-strength joints surface texturing was applied as a pretreatment

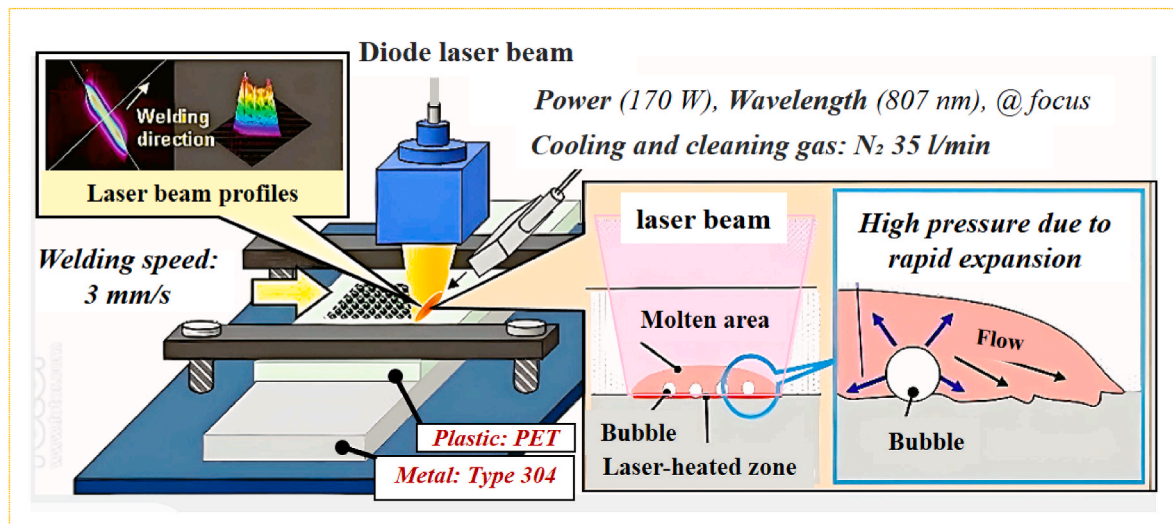


Fig. 8. Metal/polymer welding using LWP (generation of bubbles causes a high pressure that forces the melted polymer to adhere to the surface of the metal sheet) [159].

technique. The laser energy was absorbed by the Al sheet which was located in front of the laser beam above the PET sheet. Results indicated that the JS increased by increasing the LIP till a certain value at which the JS declined. Surface texturing augmented the JS due to the formation of mechanical anchorage. Both chemical and mechanical bonding mechanisms occurred during the welding of the investigated materials. The maximum obtained JS of the welded joint was 48.4 MPa, which was approximately equal to the strength of the PET sheet. Fernandes et al. [161] examined the dissimilar joining of PMMA and S235 galvanized steel using a pulsed Nd: YAG laser. The primary aim was to evaluate how processing parameters and surface conditions affected joint strength and quality. Key findings revealed that laser beam diameter and pulse duration significantly impacted joint strength. Welded samples usually failed within the PMMA component, indicating efficient joining. Mechanical pre-treatment led to a 5.1% increase in tensile shear strength. The study concluded that optimizing process parameters enabled effective joining of PMMA and S235 steel, with surface pre-treatments improving mechanical interlock. Lap welded joints between PMMA and S235 steel were created using conduction joining with an Nd:YAG laser. Optimal laser process parameters were identified, defining the maximum temperature and heating rate at the interface, which needed to stay within PMMA's limits to prevent decomposition and bubble formation. Although the tensile strength of the welded samples was lower than the base material, failures occurred in the PMMA, not the joint, due to metal-induced indentation and bending stresses. The joint strength was over 50% of the base material's strength, with failures outside the joint region, reflecting high-quality welds. Surface

pre-treatments improved direct laser joining by enhancing mechanical interlocking between PMMA and S235 galvanized steel, achieving a 5.1% increase in joint strength. However, only three types of grits were explored, indicating further opportunities for improvement. Zhang et al. [162] welded a lap joint made of A7050 Al alloy and CFRP sheets using LWP. Two types of A7050 sheets were used, namely unprocessed sheets and processed sheets. The processed sheets were subjected to a laser Surf-Sculpt process; the processed sheet is shown in Fig. 9. The shear strength of the welded joint was 8.5 and 39.0 MPa using LIP of 1200 W and BSS of 6 mm/s for unprocessed and processed sheets, respectively. The protrusion helps in enhancing the adhesion between A7050 and CFRP sheets. The JS was proportional to protrusion density. In another study [163], Al sheets were welded to GFRP sheets using the same technique used in Ref. [162]. Very high shear strength was obtained using that technique (up to 42 N/mm²). Jung et al. [164] succeeded in welding A5052 Al alloy to CFRP sheets. The highest shear strength of the welded joint was obtained at BSS of 3 mm/s and LIP of 1100 W. The fracture of the tested specimens was observed in the melted region of the CFRP or on the joint interface. EDS analysis of the joint interface indicated the existence of Al, C, O, Si, and Mg. It was reported that a nano-film containing O and Al produces a chemical bonding between the welded material. Zou et al. [165] welded CFRP composite to Ti-6Al-4V sheets using a YLS-2000TR continuous laser with a wavelength of 1069 nm installed on KUKA's KR-16 robot. Low BSS (8 mm/s) produced a joint with a lower shear strength compared with higher BSS (10 mm/s). However, increasing BSS by more than 10 mm/s resulted in a severe decrease in shear strength. A transition 2 μ m thickness film was

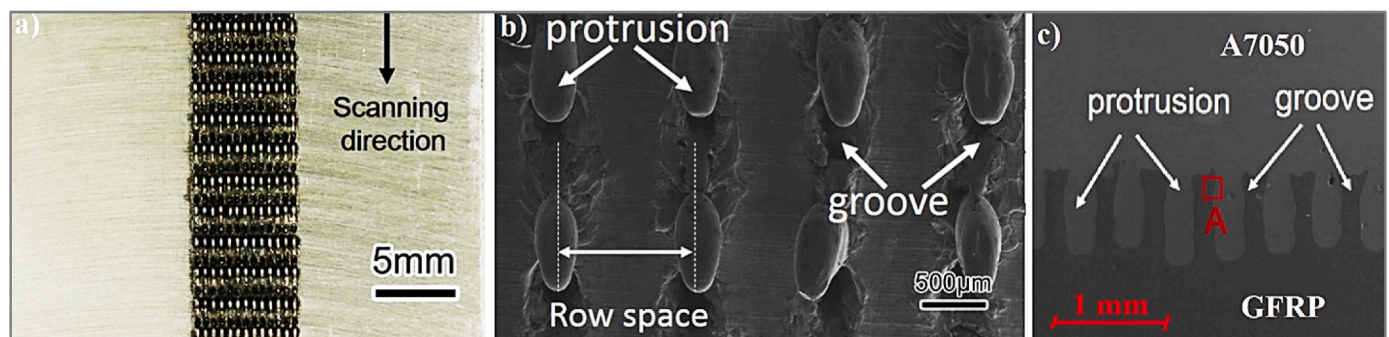


Fig. 9. Processing of A7050 sheet using Surf-Sculpt laser technique: a) A7050 sheet with protrusions; b) Microscopic image of the protrusions; c) Cross-sectional image of GFRP/processed A7050 interface after welding [162].

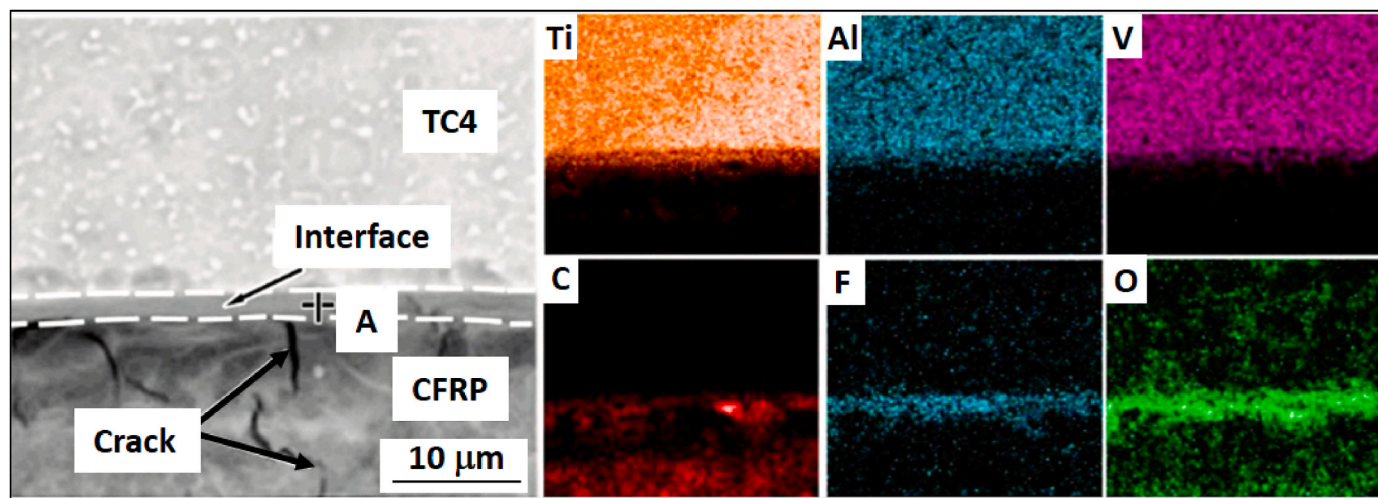


Fig. 10. The interface film with EDS element scanning [165].

formed at the interface between the welded sheets. This film contained micro-cracks. The surface analysis of the interface film using EDS showed a mutual diffusion of Ti, V, Al, F, and O elements as shown in Fig. 10. These elements precipitated in a chemical reaction at the joint interface to form chemical bonds between the welded sheets. The effects of molecular structure and JS of hybrid metallic/polymeric joints welded using the LTW technique were explored in Ref. [166]. The hybrid joining mechanism was investigated for different polymeric materials including HDPE, LDPE, PA6, PS, PET, PETG, PP, and PMP. The pressure of expansion caused by pyrolysis fragments was responsible for the bonding mechanism. They conducted experiments and analyzed the chemical composition of the welded samples to derive a quantitative relationship between the JS, molecular structure, composition gas fragment, and expansion pressure. The results confirmed the strong relationship between the JS and the structure of the welded polymer. It was also reported that JS increased by applying high pressure of expansion of pyrolysis fragments. Wu et al. [167] explored residual stresses in laser-welded CFRP and steel joints using both experimental and numerical methods, specifically employing the hole drilling method (HDM) and thermal elastic-plastic FEA. By accounting for the anisotropic properties of CFRP, the study successfully measured the non-uniform residual stress distribution through the thickness of CFRP and steel plates. Numerical simulations were performed to analyze the stress distribution at the CFRP/steel interface, focusing on stress formation mechanisms and the effects of laser welding parameters. Key findings indicated that increasing laser power led to higher longitudinal and transverse residual stresses in both CFRP and steel due to increased temperature gradients. Numerical simulations closely matched experimental results near the CFRP and steel surfaces, demonstrating the effectiveness of these methods for analyzing hybrid structures. In the steel along the weld seam, longitudinal and transverse residual stresses were tensile, with values of 514 and 91 MPa, respectively. At the CFRP/steel interface, constraints caused significant plastic strains during heating, leading to a decrease in stress magnitudes from the interface to the steel surface. Due to its low thermal conductivity and weaker matrix, CFRP exhibited lower residual stresses. During laser welding, CFRP primarily restricted the expansion and contraction of steel, resulting in asymmetrically distributed residual stresses. In another study, experiments and simulations demonstrated that laser welding effectively joined dual-phase steel DP590 and CFRP by analyzing the impact of varying laser power on joint characteristics, bonding mechanisms, and fracture behavior [168]. The study found that higher laser power increased the melting width and depth of the CFRP, achieving effective bonding between 400 and 700 W, with the highest tensile-shear peak load of 3855 N at 700 W. However, insufficient resin

melting and interfacial residual stress at lower power levels led to unbonded steel/CFRP interfaces, while excessive heat at 800 W caused CFRP decomposition. Three fracture modes were identified: interfacial fracture (IF) at 300 and 400 W, cohesion fracture (CF) in CFRP at 500 and 600 W, and a combination of interfacial and cohesion fractures (IF + CF) at 700 and 800 W. Higher laser power resulted in deeper penetration, larger melting width, and increased porosity in CFRP due to thermal expansion mismatch between the resin matrix and carbon fibers. X-ray photoelectron spectroscopy revealed the formation of C-M and O-M chemical bonds at the interface due to interactions between the resin matrix and DP590, with stronger bonds at higher laser power enhancing fracture load. Maintaining a high interfacial temperature increased the bonding area and chemical reactions at the steel/CFRP interface, improving bonding strength, but excessive temperatures could decompose the resin matrix, weakening joint strength. Therefore, peak interfacial temperatures should be maintained between 310 and 335 °C. Chan and Smith [169] used LTW to weld PET sheets to pure Ti using a fiber laser with an LIP of 200 W. It was observed that high LIP (more than 50 W) resulted in a high concentration of discoloration in the welded region regardless of the stand-off distance or BSS. This discoloration is detrimental to JS. So, it was recommended to use low LIP during LWP. Jung et al. [170] applied surface oxidation as a pre-process technique on Zn-coated steel sheet before welding it to an ABS sheet using fiber LWP. The steel sheet was preprocessed in a furnace at different temperatures of 300–600 °C for 5 min. Then, the heated specimens were cooled down at a cooling rate of 1 °C/min to room temperature (RT). A chemical bonding, containing ZnO, Zn, Fe, O, and C elements, was observed between the steel sheet and ABS. To obtain high-strength joints, it is recommended to pre-process the steel sheet at 400 °C as this pre-processing temperature in order to produce a thicker ZnO layer, which consequently enhances the JS. Schricker et al. [171] developed an innovative monitoring system composed of optical sensors to control the LWP of polymer-metal joints composed of 6082 Al sheets, AISI 304 steel, PA6, and/or PP. Fiber Bragg grating sensors have been embedded in the welding region. The effects of welding energy, material properties, and clamping pressure on the spectral response have exhibited significant influence on the spectra (distortion, peak shifting, broadening, and splitting) resulting from the induced strains and generated heat during the welding process. It was observed that high residual strains occurred at high welding energy. Arai et al. [172] employed plasma or UV–ozone as pretreatment techniques to modify the surface of COP sheets before welding to SUS304 stainless steel sheets using LWP. During UV–ozone process, the power density, wavelength, and standoff distance were 10 W/cm², 250 nm and 30 mm, respectively. During the plasma process, the gas flow rate, output power, and gas

pressure were 10 SCCM, 100–200 W and 10 Pa, respectively. The duration of the pretreatment process was considered as a process control variable. Steel sheets were also pre-processed using reactive-ion etching at a power, oxygen flow rate, gas pressure and pretreatment duration of 200 W, 10 SCCM, 10 Pa, and 300 s, respectively. The pretreatment process significantly improved the JS, however, excessive pretreatment lessened the JS.

Welding polymers to metals introduces additional complexity due to the disparate thermal and physical properties of the materials. Critical factors include the selection of a laser type that can effectively manage the heat input for both materials, such as a high-power CO₂ laser or a diode laser. The process must address the differences in melting points and thermal conductivities, often requiring specific absorbers or coatings to enhance the polymer's absorption of laser energy. Additionally, the laser parameters must be carefully optimized to prevent thermal damage to the metal and ensure a strong bond between the polymer and metal. Effective joint quality is achieved through a balanced approach to energy application and careful consideration of the material interactions at the interface.

5. Modeling

Numerical and analytical modeling of manufacturing processes are widely used techniques for predicting and investigating the relationship between the process parameters and the corresponding responses without exerting costly time-consuming experimental efforts [1, 173–176] [177]. There are several research studies carried out on the modeling of LWP that attempted to explore the weld pool evolution, heat transfer process, and the induced thermo-mechanical stress field. The optical properties along with thermo-physical properties of different polymeric materials are critical parameters that should be carefully applied during different modeling processes [178]. These properties are essential to define the amount of absorbed energy and the generated temperature field [179]. Acherjee [180] developed a coupled thermo-mechanical FEA model using ANSYS® to analyze temperature fields and stress distributions during the heating and cooling of polymer LWPs. The model incorporated temperature-dependent thermo-mechanical properties and simulated heat transfer via radiation, conduction, and convection as shown in Fig. 11. It used the von Mises yield model with isotropic hardening and Perzyna viscoplasticity to capture

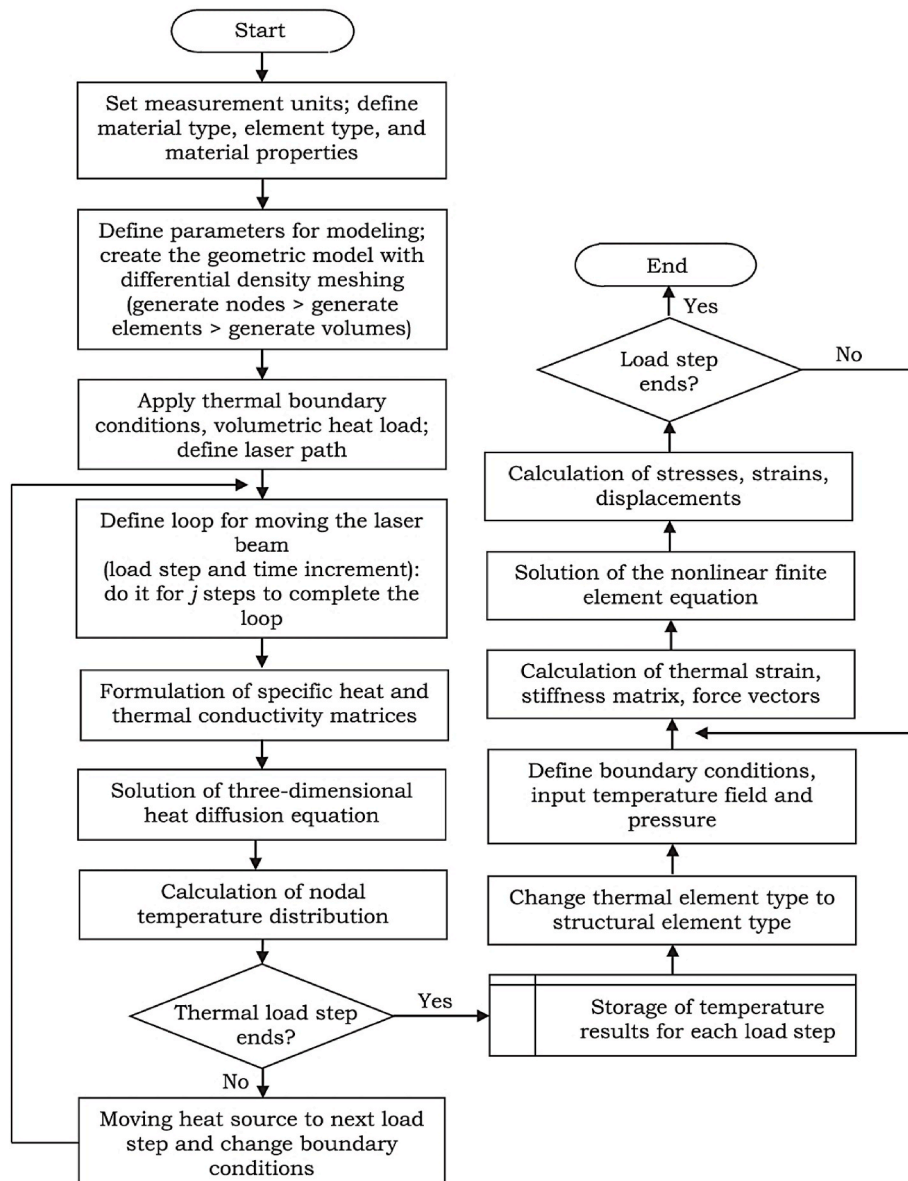


Fig. 11. Flowchart of coupled thermo-mechanical FE model of LWP [180].

material behavior. The study found that maximum stress occurred near the beam position and weld pool, with compressive stresses in the heat-affected zone during heating shifting to tensile stresses during cooling. In another study [181], the same author simulated the heat transfer process during LTW considering the quasi-simultaneous LWP using ANSYS®. The effects of process variables namely BSS, LIP, and welding passes on the induced temperature field were investigated. Higher LIP and BSS resulted in higher temperatures in the welding region. It was also reported that WSW increased with LIP and the number of passes, while it decreased at higher BSS. Another quasi-simultaneous LWP model was developed using COMSOL Multiphysics to simulate the LWP of PP [93]. It was reported that the highest absorption was observed in the upper welded component because of the scattering characteristics of PP. COMSOL Multiphysics was also used to simulate the quasi-simultaneous LTW of PC [49]. Narrow HAZ was obtained by applying a higher number of passes and feed rates. ABAQUS software was also employed to simulate the LWP of PBT GF-ASA and PC considering quasi-simultaneous welding conditions [182]. The FE model was used to investigate the effect of temperature on the JS. The correlation analysis carried out on energy input and welding temperature indicated that the JS was highly affected by welding temperature followed by the energy input. Nevertheless, the energy input has also a considerable correlation to the weld strength. Hopmann et al. [183] developed an FE model using ABAQUS to simulate residual stresses induced in polymeric materials welded by the LTW technique. These residual stresses resulted from the thermal expansion of welded materials under subsequent heating and cooling cycles. To compute the induced residual stresses, LTW was modeled as a three-dimensional (3D) heat transfer process by employing Fourier's law of heat conduction on PA66 material. Three different behaviors of the welded material were considered in the developed model, namely: elastic, linear viscoelastic, and elastoplastic. It was found that the elastic-plastic model outperformed other models to compute thermally generated residual stresses. Acherjee [112] developed a 3D model using ANSYS® software to simulate the LTW of PC to ABS sheets. The parametric design language was employed to model a moving volumetric heat source that had a Gaussian nature. Thermophysical properties of the welded materials were considered temperature-dependent properties. Both convection and radiation heat loss to the surroundings were considered in the developed model. The dilution effect was also considered to determine the weld pool composition as two different materials with disparate chemical composition and thermo-physical properties were welded. It was reported that the bottom absorbing material (ABS sheet containing CB) had higher temperatures than not only the upper PC sheet but also the weld interface. The widths of melted regions at the weld interface for both materials are not identical because of the variation in the glass transition temperatures between ABS and PC. Acherjee et al. [184] numerically investigated the importance of CB on temperature field and weld geometry during LTW of polymers. They developed a time-dependent heat transfer model of the welding process on ANSYS®. The heat source was considered a Gaussian-type volumetric heat source. The model was applied to welded PC sheets with different concentrations of CB. The obtained results revealed that CB has a significant effect on the temperature field and the weld geometry. At high CB concentrations, the geometry of the joint becomes symmetrical as the high CB concentrations caused surface absorption instead of volumetric absorption. Chen et al. [185] developed a quasi-static 3D FE model to simulate the heat transfer process considering mass flow during LWP of PC. The model showed a good agreement with the experimental data with minimal computational cost. The simulation results indicated that the maximum observed temperature increased with increasing BSS at constant line energy. A 3D FE model was developed using ANSYS software to simulate the time-dependent heat transfer process during spot LTW of nylon 6 sheets [186]. The front and top faces of the welded components were subjected to convection heat transfer, while the remaining faces were considered insulated surfaces. A MATLAB code

was established to represent heat generation input as a function of LIP, LS dimensions, and optical properties of the welded components. Chen et al. [187] developed an analytical model to simulate LWP considering the scattering behavior of laser beams in welded sheets. The scattered energy has been represented by Gaussian distribution. The energy transmitted in welded was described by reflectance, transmittance and absorbance. The developed model was based on two main parameters, namely standard deviation of scattering and the scattering ratio. The model was employed on PP, PA6 and GF-PA6 and exhibited reasonable accuracy compared with experimental results. Zoubair and Elhem [188] developed an FE model in ABAQUS to determine the residual stress distribution and temperature field generated in a PP tank after LWP and the technique employed was laser diode transmission welding. They established a FE model considering different thermal and mechanical conditions such as heat source properties, material thermo-mechanical properties, and applied pressure. The model considered temperature-dependent material properties. The following LWP parameters were applied during the conducted simulations: a constant BSS equal to 110 mm/s, a constant LS with a diameter of 2 mm, a constant LIP of 100 W and CP varies between 0.3 and 0.5 MPa. It was recommended to use low CP to obtain improved temperature, stress, and strain distributions (see Fig. 12) as well as a low rigid bending effect. Acherjee et al. [189,190] modeled and analyzed the impact of key process parameters—beam diameter, power, welding speed, and CB content—on the weld width, temperature, and depth in laser transmission contour welding of PC by employing ANSYS® multi-physics along with a simple design of experiments approach. Mayboudi et al. [191] developed an FV model to explore the thermal response of a laser-absorbing agent (CB particles) to LWP of T-joint made of PA6. The laser was set at LIP of 60 W power and BSS of 10 mm/s. The obtained results were compared with those obtained from the FE model developed using ANSYS®. There was a good agreement between the results of both models. The temperature field and radiation propagation were simulated for LTW of PC and PBT sheets using COMSOL Multiphysics [192]. The simulations were conducted for an M-shape and Gaussian-shape laser beam. The laser beam was scattered into the PBT sheet more than that of the PC sheet. This resulted in destroying the original shape of the employed beam. For the PC sheet, the seam thickness in the case of a Gaussian-shape beam has a non-homogeneous nature contrary to the case of the M-shape beam as the thickness has a homogeneous nature. Gaussian-shaped beams resulted in high-temperature fields and high degradation probability. Thus, the M-shape beam is recommended to obtain good seam properties. Bates et al. [72] modeled the thermal degradation that occurs during LTW of PA6 and PC sheets as a function of time and temperature using a FE technique. Thermal gravimetric analysis was employed to determine the kinetic triplets (reaction model, activation energy and frequency factor) of the welded materials by applying a non-linear fitting model. The temperature field as a function of time was computed using the FE model and was combined with the computed kinetic triplets to predict material degradation. In another study, Chen et al. [193] proposed a non-contact approach to determine the BSS and LIP required for optimal contour in LTW as a function of laser characteristics and material properties. The proposed approach employs a rapid scan across the welded components using a range of BSS and LIP. The BSS and LIP required to start degradation in the absorbent component are related to the welding conditions that produce maximum shear strength. The approach has been applied to four different polymers, namely, PP, PC, PA6, and GF-PA6. It was observed that the JS increased with increasing LIP. Higher LIP induced more absorbed energy in the welding region which increased the welding temperatures and molecular diffusion between the welded components. At a certain BSS-dependent LIP, the strength decreased by increasing LIP due to the occurrence of molecular degradation. The molten area formed during LTW of PA66 containing GF or CF was predicted using a well-established analytical model [194]. The heat formula used to derive the mathematical model was based on thermotics. Adiabatic conditions were

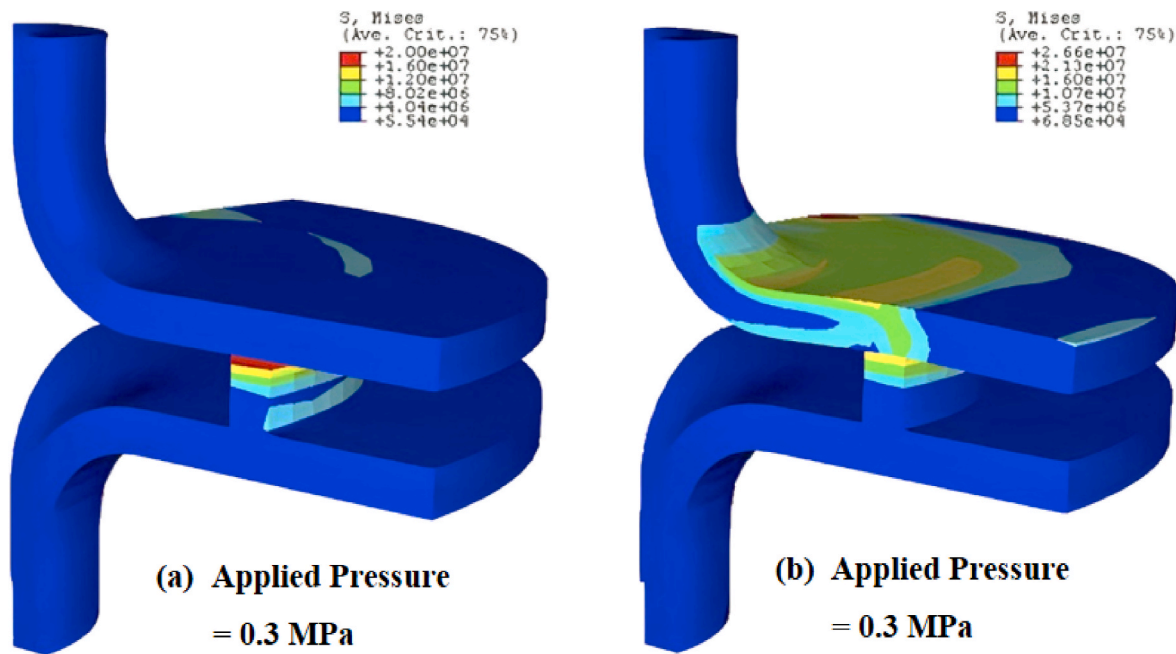


Fig. 12. Stresses distribution at different CP: a) 0.3 MPa; b) 0.5 MPa [188].

considered for temperature rise and energy absorption processes. The thermophysical properties of the welded materials were assumed to be independent of welding temperature. The effects of welding parameters such as LIP and BSS on the molten pool geometry were explored. The molten area is inversely proportional to BSS and directly proportional to LIP. Statistical approaches have been also used to model and optimize different LWP. A sequential optimization approach combining the RSM, Taguchi method, and desirability function analysis was proposed for optimizing LTW parameters [195]. The study focused on weld strength and width and evaluated the effects of welding speed, laser power, and focal position. The Taguchi method identified optimal control factors, which are then refined using RSM models developed through central composite design. Desirability function analysis further optimizes the parameters for multiple objectives, significantly improving weld quality. The proposed method outperformed the Taguchi and grey-Taguchi methods based on comparison results. Acherjee et al. [109] used the Taguchi method along with grey relational analysis to obtain the optimal welding conditions during LTW of pure PMMA and CB-enriched PMMA. Grey relational analysis was employed to change various quality characteristics into a single characteristic. This single quality characteristic is represented by the grey relational grade index which is used to obtain the optimal welding conditions. Nakhaei et al. [113] employed RSM to predict JS of laser-welded butt joints made of PP reinforced with CB and clay with different concentrations. The control factors of the established model using RSM were BSS, LIP, clay concentration and focal distance. The model predicted the weld strength with a high correlation coefficient of 0.987. RSM technique has been also successfully applied to develop a second-order model that describes the weld strength of welded ABS-PC sheets using LTW [111]. The developed model was checked for its adequacy with a confidence range of 95 %. Acherjee et al. [196] investigated the effects of key process parameters—laser beam size, welding speed, laser power, and clamp pressure—on lap-shear strength and weld-seam width during LTW of PMMA with a diode laser system. They used RSM to develop mathematical models that describe the relationship between these parameters and output variables, helping to identify optimal welding conditions. Interaction effects were analyzed using response surface plots, and the Design-Expert v7 software created the design matrix and derived regression equations. ANOVA verified model adequacy, showing good agreement between

measured and predicted results. The study concluded that increasing laser power enhances lap-shear strength and weld-seam width, while higher welding speeds reduce them. An optimal line energy level for each power setting maximizes lap-shear strength, with weld-seam width increasing until it reaches a limit. Lap-shear strength initially increases with stand-off distance but decreases beyond a certain point, while weld-seam width rises with increasing stand-off distance. Clamp pressure has a slight positive effect on lap-shear strength and an insignificant impact on weld-seam width. ANOVA results indicate welding speed significantly influences lap-shear strength, while both welding speed and stand-off distance affect weld-seam width. Validation experiments confirmed model accuracy within the specified parameter range, demonstrating reliability for optimizing welding conditions. In another study, Acherjee et al. [197] focused on diode LTW of dissimilar thermoplastics, specifically ABS and PMMA, using RSM to analyze how stand-off distance, welding speed, laser power, and clamp pressure affect weld strength and width. They developed mathematical models to correlate process parameters with responses, verified through tests like the sequential F-test and ANOVA. The study used numerical optimization with RSM for optimal solutions and graphical optimization to identify conditions visually. Results showed models accurately predict weld strength and width within tested ranges. Laser power, welding speed, and stand-off distance significantly influence weld quality by controlling heat input; excessive heat leads to decomposition, while insufficient heat causes poor fusion. ANOVA indicates stand-off distance has the greatest impact on weld strength, followed by laser power and welding speed, with clamp pressure insignificant. Optimal conditions are determined using numerical optimization, providing Pareto-optimal solutions, and graphical optimization offers visual tools for selecting the best parameter combinations.

In laser welding of polymeric materials, both analytical and numerical modeling tools face distinct challenges that impact accuracy. Analytical models often simplify boundary conditions to manage mathematical complexity, which can compromise prediction accuracy by failing to capture intricate thermal gradients, variable material properties, and complex geometries. These approximations may lead to discrepancies between theoretical predictions and actual weld outcomes. To improve accuracy, it is essential to refine boundary conditions by incorporating realistic representations of heat transfer, material

interactions, and geometric details. Advanced analytical techniques, such as perturbation methods, combined with empirical data for model validation, can enhance model fidelity.

Numerical models, on the other hand, struggle with simulating the complex thermal dynamics, phase transitions, and material behavior inherent in the welding process. Challenges include capturing the non-linear, temperature-dependent properties of polymers, such as varying thermal conductivity and viscosity, and accurately modeling the heat-affected zone and potential material degradation from laser exposure. To enhance model accuracy, incorporating temperature-dependent material properties, using adaptive meshing techniques, and integrating real-time experimental data are crucial. Multi-physics approaches that account for thermal, mechanical, and chemical interactions can provide a more comprehensive understanding of the welding process. Leveraging advanced computational resources and techniques, such as parallel processing and high-performance computing, can support the complex simulations needed for the accurate modeling of polymeric welding processes. Combining these strategies can lead to more precise analytical and numerical models that better reflect the true behavior of polymeric materials during laser welding.

6. Advanced modeling techniques

Artificial neural network (ANN) is a powerful modeling tool that can learn and find relationships between different process variables and responses by employing a training process based on experimental data. ANNs have been widely applied to various engineering fields. They have been successfully applied in modeling different processing techniques of polymeric materials such as laser cutting [198], FSW [199,200], friction stir processing [201], and ultrasonic welding [30,202]. Jeng et al. [203] predicted the parameters of LWP using ANN models. Two learning approaches were employed, namely learning vector quantization and back-propagation algorithms. The input parameters of the models were welding gap, LIP, BSS, and sheet thickness, while the output parameters were WSW, undercut, and distortion. The results of the model predictions were in good agreement with the experimental results and have the capability to be applied in the laser welding industry. Acherjee et al. [204] applied an ANN model to predict the JS and WSW of PMMA welded sheets using the LTW technique. The inputs of the model were LIP, BSS, standoff distance, and CP. Different model architectures with different numbers of hidden layers and different numbers of hidden layers were tested to obtain the optimal one. Two hidden layers with four and five hidden neurons for the first and second layers, respectively, were the optimal architecture. They also compared the predicted results of the ANN model with those obtained by multiple regression techniques considering first and second-order polynomial models. ANN model outperformed other regression models for both investigated responses. Mehrpouya et al. [138] employed two different ANN models to predict the joint characteristics of LTW pure PET sheet to PET/PEVA blends as a function of welding parameters. Two different ANN models were established for this purpose, namely radial basis function (RBF) and generalized feed-forward multi-layer perceptrons (GFF-MLP). The inputs of the models were LIP and BSS; while the outputs were elongation and maximum load of the welded joint. GFF-MLP outperformed RBF in terms of prediction accuracy of the welded joint characteristics. GFF-MLP had a fitting of over 97 and 98% for cross-validation and training values, respectively, while they were 94 and 96% in the case of RBF. Wang et al. [130] applied an ANN model to predict JS and WSW of PC welded sheets using the LTW technique as a function of LIP, BSS, scanning number, and CP. They applied the desirability function technique to convert multiple response models into a single response model. Then, the established desirability function was optimized using the genetic algorithm to maximize the joint quality. The developed model succeeded in obtaining optimal welding parameters that augment the JS to reach a high value of 43.14 MPa and minimize the WSW to reach 3578.2 μm while preserving the BSS at reasonable values. The predicted

WSW and shear strength are in good agreement with the experimental values with an error range of 4.10 and 5.23%, respectively. ANN algorithms were employed to augment the weld quality of PC/PMMA joints [141]. These algorithms are desirability function optimizers and teaching learning-based optimizers. They were used to obtain the optimal process parameters for minimizing WSW and maximizing weld shear strength. The RSM was employed in that work to design the experimental plan and to develop an empirical model that is used in the optimization process. Confirmation experiments confirmed that teaching learning-based optimizers outperformed desirability function optimizers. The teaching learning-based optimizer was also employed to optimize the LTW of PMMA sheets [131]. It was reported that the optimizer had very low computational costs for solving hard optimization problems. Wang et al. [205] utilized CB to enhance the absorption of laser energy and its conversion into heat during LWP of PC and PA6. NIH image processing package was utilized to analyze TEM images of the CB. The aggregation of CB particles had ellipsoidal shapes in both welded materials with bimodal distribution for PA6 and normal distribution for PC. Based on the reviewed literature, ANN tools as well as metaheuristic optimizers could be developed to predict and optimize different LWP of different polymeric materials. ANN tools act as black boxes which learn the relationship between the inputs and outputs via a training process using experimental data. This will help in simplifying the modeling process by avoiding solving nonlinear mathematical models using complex mathematical tools. Moreover, the nonlinear temperature-dependent materials properties will be inherently involved in the training process as the model will learn by examples. Thus, the use of ANN tools is recommended to model the LWP of polymeric materials.

Numerical methods, such as FEA, are extensively employed to model and simulate LWPs, offering detailed insights into thermal profiles, stress distributions, and material behavior throughout welding. These methods are advantageous for predicting weld quality and performance under various conditions, allowing for an in-depth understanding of temperature distribution, phase transitions, and potential thermal damage. By optimizing process parameters through simulations before conducting physical experiments, FEA can significantly save time and resources. However, challenges include the necessity for accurate material properties and boundary conditions, which are often difficult to obtain. Additionally, numerical simulations can be computationally intensive and may involve simplifying assumptions that can affect prediction accuracy.

In contrast, analytical methods involve solving mathematical equations that describe the welding process, typically focusing on simplified models of heat transfer and material behavior. These methods are advantageous for their relatively straightforward implementation and the ability to derive general principles and relationships between process parameters and weld quality. They provide quick insights and are less computationally demanding compared to numerical simulations. Nonetheless, the simplifications necessary to make equations solvable may not fully capture the complexities of real welding processes, potentially leading to less accurate predictions in complex scenarios.

Statistical methods, such as regression analysis and design of experiments (DOE), are used to explore the relationships between welding parameters and weld quality. These methods excel in identifying significant factors and interactions through empirical data analysis, helping to optimize process parameters and predict outcomes based on historical data. However, their reliance on empirical data can sometimes miss out on all influencing factors, and there is a risk of overfitting or underfitting the models if the data is not sufficiently comprehensive or representative.

AI techniques, including machine learning and ANN, provide advanced capabilities for predicting and optimizing welding processes. The advantages of AI include its ability to handle complex, non-linear relationships between parameters and weld quality and to improve prediction accuracy by learning from large datasets. AI models can adapt to new data and uncover patterns that traditional methods might

overlook. However, AI approaches require extensive high-quality data for effective model training and can act as black boxes with limited interpretability. They also demand significant computational resources and require rigorous validation and testing to ensure their reliability and applicability in real-world scenarios.

7. Recent advancements in laser welding for polymers

Laser welding offers numerous benefits across various industries, including automotive, aerospace, and medical device manufacturing [206]. In the automotive industry, laser welding is used to join lightweight polymer components, aiding in weight reduction and improving fuel efficiency while enabling the creation of complex shapes and joints that traditional methods struggle to achieve. In aerospace, laser welding creates strong, lightweight polymer components and allows for joining dissimilar materials, enhancing design possibilities and performance. The precision and cleanliness of laser welding make it ideal for medical devices, where biocompatibility and sterility are critical, enabling intricate designs and the fabrication of micro-scale components [207].

Different polymers present diverse thermal properties, absorption characteristics, and mechanical behaviors, which can complicate the LWP and impact weld quality. Polymers with high thermal conductivity or varying melting points may require customized welding parameters to achieve optimal results. Additionally, some polymers might degrade or emit harmful gases when exposed to laser energy, which demands advanced process controls and protective measures [208]. Thus, this paper delve into these material-specific challenges and suggest solutions or adjustments to accommodate the wide range of polymer types. Scalability is another critical aspect, especially for industrial applications. Transitioning from laboratory or small-scale setups to full-scale production can be challenging, as it involves ensuring consistent weld quality and managing thermal effects over larger areas. Future research should focus on innovations in automated systems, real-time monitoring, and adaptive control technologies to overcome these scalability issues. Integrating laser welding with emerging technologies like advanced robotics, AI-driven process optimization, and new polymer formulations could further enhance precision, efficiency, and versatility in welding applications. Addressing these challenges and exploring these future research avenues would provide a comprehensive understanding of the current and evolving landscape of laser welding technology for polymers.

Recent advancements in laser welding technology for polymers and absorber materials have indeed transformed the landscape of plastic and polymer welding. As industries like automotive, aerospace and medical devices increasingly demand lightweight, durable, and complex polymer structures, innovations in laser welding have become essential for addressing these challenges. In this section, an overview of the key advancements in laser welding technology for polymers will be discussed:

Fiber lasers, diode lasers, and ultrashort pulse lasers are key technologies in polymer welding, each offering unique advantages in beam quality, stability, and efficiency [209–211]. Fiber lasers are favored for their precise control, allowing for high-speed welding with minimal heat-affected zones, while diode lasers are valued for their compact size and high-power densities, making them ideal for integration into automated systems for mass production [212]. Ultrashort pulse lasers, which emit pulses in the picosecond or femtosecond range, are particularly suited for precision welding with minimal thermal damage, making them especially useful for welding thin and delicate polymer films [213, 214]. Additionally, advanced optics and beam shaping techniques [215], such as beam splitting and beam oscillation, enhance the flexibility and precision of the welding process by enabling tailored energy distribution, which is critical for joining complex geometries.

Absorptive dyes, pigments, and nanoparticles are crucial in the laser welding of polymers, as they absorb laser energy and convert it into heat, facilitating stronger and more consistent welds [137,216]. Recent advancements in absorptive dyes and pigments have enhanced energy

absorption, resulting in improved weld quality [217]. Nanoparticle absorbers, such as CB and metal oxides, are being incorporated into polymers to boost laser absorption and provide better control over the welding process due to their uniform dispersion in the material [117]. Additionally, innovations in clear polymer welding have led to the development of specialized absorbers that allow laser transmission through one component while being absorbed by another, enabling the welding of transparent or translucent polymers without leaving visible marks on the surface.

Precision and control in laser welding have been significantly enhanced through the integration of real-time monitoring systems and the application of machine learning and AI [218]. Real-time monitoring systems, such as thermal cameras and sensors, provide immediate feedback and enable adaptive control, ensuring consistent weld quality by allowing precise control over the welding process. Additionally, AI and machine learning algorithms are increasingly being used to optimize welding parameters and predict outcomes, further improving the precision and efficiency of LWP [219].

Challenges in laser welding include material compatibility, thermal management, and sustainability, each presenting unique opportunities for future advancements. Developing absorber materials compatible with a broader range of polymers is an ongoing challenge, with research focused on creating versatile absorbers. Effective thermal management is crucial to prevent damage to sensitive components, necessitating advancements in cooling techniques and heat management systems to expand laser welding applications. Additionally, as industries shift towards more sustainable practices, laser welding technology is evolving to reduce energy consumption and improve the recyclability of welded components, aligning with environmental goals and enhancing the overall sustainability of manufacturing processes.

8. Summary and conclusions

This review explores laser welding technology, a prevalent fusion welding method for joining polymeric materials using laser radiation as the heat source. In this process, laser energy is absorbed by an opaque polymeric material and converted into heat. The heat generated and the temperature distribution during welding can be controlled by adjusting parameters such as laser power, scan speed, number of scans, and laser diameter. Identifying the optimal welding parameters is crucial for enhancing weld quality in terms of joint strength and seam width, given the nonlinear behavior of polymers under heat. Using too low a laser power can result in welding failure, while excessive power may burn the materials due to overheating. Additionally, high welding speeds necessitate higher laser power, and the optimal speed is influenced by both the laser power and the material's optical and thermophysical properties, such as absorption coefficient and thermal diffusivity. Joint quality is more dependent on the temperature field than on energy input. For multi-scan welding, it is important to select the scan distance to prevent the re-solidification of the welding pool before the process completes. Proper clamping of the welded parts is crucial for maintaining contact between components and improving material mixing in the weld pool. Analytical modeling of the LWP is complex and often involves simplified boundary conditions that can compromise accuracy. Conversely, numerical modeling using FEA methods entails high computational costs. AI has proven effective in modeling and optimizing the welding process due to its ability to learn and generalize.

Future trends in laser welding of polymeric materials are set to enhance process efficiency, broaden material compatibility, and improve joint quality. Developments include high-precision lasers with refined beam control to reduce thermal degradation and increase weld accuracy. Research is focusing on novel polymer formulations and additives to boost weldability and bond strength. Innovations aim to enhance welding speed and energy efficiency, with real-time monitoring and adaptive control systems being integrated to ensure optimal weld quality. There is also a rising interest in hybrid welding techniques that

combine lasers with other methods to effectively join complex or dissimilar materials, expanding laser welding's applications in high-performance polymer joints. Future studies should investigate the impact of various pulse regimes on weld quality and explore advanced absorber-free or low-cost absorber-based welding technologies for different polymers.

Declaration of competing interest

The authors of the submitted manuscript **"Recent Advances and Future Prospects of Laser Welding Technology for Polymeric Materials: A Review"** declared that there is no conflict of interest. All authors mutually agree this manuscript to be submitted to **Journal of Materials Research and Technology**, which was not previously submitted to this journal. Moreover, this paper is not being submitted to any other journals.

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